AN OFFLINE, BEGINNER-FRIENDLY COMPANION

Can the hotter start finish closer?

Walk through the four-state cooling research without assuming a course in statistical mechanics or linear algebra. We build the vocabulary, do the arithmetic, and separate what is proved from what a graph merely suggests.

> **A gradual course with an inspectable evidence trail.** Start with the small examples and work toward the proofs. The current report has detailed statement reviews, independent calculations, reproduction receipts, and a checked Lean companion. Step 20 explains exactly what those checks cover.

Keep this file in the project root so evidence links work. Everything needed for the interactive lessons is embedded; no network or account is required. Calculators use ordinary browser arithmetic and are illustrations, not proof certificates.

Read the expanded report: 40 gradual lessons followed by the complete technical report (126-page PDF). Start with fractions and probability; continue through worked matrix calculations, the counterexample, advanced proofs, and 14 exercises with answers.

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STEP 01 · THE QUESTION

Understand the claim before checking it

Imagine two preparations of the same system. One begins hotter than the other. Put both in contact with the same cooler bath and let both evolve under the same rules. Usually the hotter preparation starts farther from the bath's equilibrium. The question is whether it can later be *closer*.

Here “closer” means a specified mathematical distance between probability distributions. It need not mean a lower thermometer reading. The main example has four possible states. A proposed shortcut inspects every group of three states and tries to rule out unusual cooling. The example shows that this shortcut can miss a collective effect.

1. Verify that the four-state model is a valid thermalizing model.
2. Apply exactly the stated three-state tests to all four triplets.
3. Verify that the hotter preparation begins farther away.
4. Prove that it eventually becomes closer.

A **counterexample** is one valid case contradicting a universal statement. It does not say that every system behaves that way. This counterexample concerns a finite-temperature conjecture; it does not refute the separate zero-temperature theorem.

▼ **Checkpoint: What would not be enough?**

A plot with a crossing, without checked inputs and error bounds, would not establish the full claim. Nor would a passing triplet test on only three of the four triplets.

STEP 02 · PROBABILITY

Think of four boxes holding one unit in total

The numbers p_1, p_2, p_3, p_4 describe how likely the system is to be in each state. Each lies between zero and one, and together they sum to one. A vector is simply an ordered list. We write the probabilities in a column.

```
p = (0.4, 0.3, 0.2, 0.1)
0.4 + 0.3 + 0.2 + 0.1 = 1
```

These probabilities can also describe the fractions in a very large collection of identical systems. They do not mean that a single classical system is partly in each state.

If a state is assigned a value w_i, its probability-weighted average is $\Sigma_i w_i p_i$. The symbol Σ means "add the following terms." For values (1, –1, 0, 0), the average in the example is 0.4 – 0.3 = 0.1. Negative values of w are allowed: w is an observable, not a probability.

▼ **Try it: average (2, 0, –1, 3)**

2(0.4) + 0(0.3) – 1(0.2) + 3(0.1) = 0.9. We will use exactly this operation to calculate a slow-mode

overlap.

STEP 03 · THE GIBBS DISTRIBUTION

Higher energy is less likely at equilibrium

Assign an energy E_i to each state. At temperature T, give it an unnormalized weight $\exp(-\beta E_i)$, where $\beta = 1/(k_B T)$. We choose units with $k_B = 1$ and bath $\beta_b = 1$. The exponential $\exp(x)$ is e^x; its inverse is the natural logarithm ln.

$$Z(\beta) = \exp(-\beta E_1) + \exp(-\beta E_2) + \exp(-\beta E_3) + \exp(-\beta E_4)$$
$$\pi_i(\beta) = \exp(-\beta E_i) / Z(\beta)$$

Z is just the normalizing total. The Greek letter π here names the equilibrium probability list; it is not the circle constant. **A smaller β means a hotter preparation.** At β near zero the probabilities are nearly equal. Cooling increases β and favors low-energy states.

It is convenient to store the bath weights $q_i = \exp(-E_i)$. Then $E_i = -\ln(q_i)$, and at any preparation temperature the weights are q_i^{β}. In this project the bath remains fixed while the preparation changes.

▼ **Checkpoint: Which is hotter, β = 0.001 or β = 0.01?**

β = 0.001 is hotter. In bath-temperature units their temperatures are 1000 and 100. Both are hotter than the bath at T = 1. These very hot preparations are part of a mathematical counterexample, not the later moderate-range physical model.

STEP 04 · RATES

Probability flows between boxes

A transition rate is a probability per unit time, to first order over a very short interval. Rate 2 does not mean probability 2. Over a short time Δt, it gives a transition probability approximately $2\Delta t$. Multiple transitions make this approximation unsuitable for a large Δt.

M_{ij} is the rate **from state j to state i**. Think "column gives the starting box, row gives the destination." The diagonal records loss, so it is negative.

$$M_{jj} = -\sum_{i \neq j} M_{ij}$$

$$dp_i/dt = \sum_j M_{ij}\, p_j$$

Each column of M sums to zero.

A matrix is a table of numbers. Multiplying M by p means taking each row's weighted sum. The equation above is a collection of ordinary differential equations. Column sums of zero make the total derivative of $p_1 + p_2 + p_3 + p_4$ zero. Positive off-diagonal rates ensure that probability cannot flow out of an empty box.

▼ **A complete two-box example**

$$M = \begin{bmatrix} -2 & 1 \\ 2 & -1 \end{bmatrix}$$

$$dp_1/dt = -2p_1 + p_2$$

$$dp_2/dt = 2p_1 - p_2$$

Add the equations to get zero. The equilibrium is (1/3, 2/3): the outgoing flow 2(1/3) equals the returning flow 1(2/3).

STEP 05 · EQUILIBRIUM

Check every forward-and-backward pair

Detailed balance means that at the bath distribution, each pair has equal flows in both directions:

$$M_{ij}\, q_j = M_{ji}\, q_i.$$

Dividing both sides by $Q = q_1 + q_2 + q_3 + q_4$ gives equal flows for the actual probabilities. There may still be individual transitions at equilibrium; their average flows cancel.

For $i < j$, let a_{ij} be the downward rate from higher-energy j to lower-energy i. To impose detailed balance, define the upward rate as $a_{ji} = a_{ij}\, q_j / q_i$. Since $q_j < q_i$, going up costs energy and is slower.

Example: $q_1 = 1$, $q_2 = 199/200$, $a_{12} = 1$

$a_{21} = 1 \times (199/200)/1 = 199/200$

$a_{12}\, q_2 = 199/200 = a_{21}\, q_1.$

Check all six pairs and all four column sums. Together these give $M\pi_b = 0$: the bath distribution stays fixed.

STEP 06 · EXACT INPUTS

Build the focused paper's actual model

$q = (1, 199/200, 2/5, 1/400)$

$E = (0, \ln(200/199), \ln(5/2), \ln(400))$

$Q = 959/400$

$\pi_b = (400, 398, 160, 1)/959$

The energies are approximately (0, 0.005013, 0.916291, 5.991465). They are strictly increasing. Use these six downward rates:

From → to	Exact rate
2 → 1	1
3 → 1	28
4 → 1	249/10
3 → 2	330154675693/1470375000
4 → 2	4983/200
4 → 3	1527327066/151990625

Fill the other six entries using detailed balance, then compute the four diagonal losses. For example, the first diagonal is −[199/200 + 28(2/5) + (249/10)(1/400)] = −12.25725 exactly.

> **Do not mix witnesses.** The comprehensive report also has an earlier “primary” example with slightly different rates. Its overlap percentage and certified time brackets do not automatically apply to this focused-paper example. The standalone primary checker and the new walkthrough checker test different generators.

Inspect the resulting full matrix in the new exact-arithmetic receipt.

STEP 07 · FOUR TRIPLETS

Apply the original test without changing it

The four ordered triplets are 123, 124, 134, and 234. For each $i < j < k$, set $A = a_{ik}$, $B = a_{jk}$, $C = a_{ij}$, $t = q_j/q_i$, and $u = q_k/q_j$. The letters t and u in this test are weight ratios, not time.

```
κ = (A+B)/[C(1+t)]
G = At − B
F = A(1+tu/2) − B(1+u/2) − C(1−t)
P = (κ ≥ 1 AND F > 0) OR (κ ≤ 1 AND G > 0)
```

“P is false” means that this necessary-condition test fails. It is not a claim that every passing triplet necessarily has an effect. Notice that κ comparisons include equality, whereas the F and G comparisons do not.

Do triplet 124 by hand

```
A = 24.9, B = 24.915, C = 1
t = 199/200, u = 1/398
G = (249/10)(199/200) − 4983/200 = −279/2000
κ = 49.815/1.995 > 1
F = −0.020175251256… < −0.02
```

Only the κ ≥ 1 branch can pass here. But F is negative, so the triplet test fails. Exact arithmetic gives the following bounds for all four:

Triplet	Branch	F	G
123	κ > 1	< −230	< −196
124	κ > 1	< −1/50	< −13/100
134	κ < 1	< −19/10	< −2/25
234	κ < 1	< −118	< −3/100

▼ **Why use original rates?**

The conjecture specifies rates from the original system. Eliminating a state and replacing its routes by “effective rates” changes the test. If you form an isolated triplet generator, recompute its diagonal losses;

deleting a row and column alone leaks probability.

STEP 08 · LOGIC AND ALGEBRA

Check the useful shortcut in all three cases

The test fails exactly when both $F \le 0$ and $G \le 0$. This is not true of arbitrary pairs of numbers with arbitrary branches; it follows from a special identity.

$$H = A+B-C(1+t),\ d = (1-t)/(1+t),\ h = 2/(1+t)+u/2$$
$$F = hG + dH, \text{ with } h > 0 \text{ and } d > 0.$$

1. **$H > 0$:** $\kappa > 1$, so failure means $F \le 0$. Rearranging gives $hG = F - dH < 0$. Thus $G < 0$.
2. **$H < 0$:** $\kappa < 1$, so failure means $G \le 0$. Then $F = hG + dH < 0$.
3. **$H = 0$:** $\kappa = 1$. Both branches apply, and $F = hG$. Failure means neither strict sign is positive, so F and G are both nonpositive.

Conversely, if both are nonpositive, neither branch can pass. At fixed q, each F and G is linear in the downward rates, so the allowed rates form an intersection of half-spaces. A half-space is simply one side of a flat boundary. Equalities belong to the allowed set.

▼ **Checkpoint: If $\kappa = 1$ and $F = G = 0$, does the test pass?**

No. The sign tests require strict positivity. This matters in the limiting family used for the 25% theorem.

STEP 09 · EXPONENTIAL DECAY

A mode is a pattern that keeps its shape

First recall the calculus equation $y' = \lambda y$. Its solution is $y(t) = y(0)e^{\lambda t}$. If λ is negative, the pattern decays. If $\lambda = -2$, it decays more slowly than $\lambda = -10$.

For several boxes, a **right eigenvector** v is a pattern satisfying $Mv = \lambda v$. Starting with that pattern's deviation from equilibrium, the deviation remains proportional to v and its size changes by $e^{\lambda t}$. A **left eigenvector** w satisfies $w^{\mathsf{T}}M = \lambda w^{\mathsf{T}}$. It measures how much of that mode is present.

$$w = (1, -759/1000, -48957/80000, -1/250)$$
$$\lambda_2 = -4966671/250000 = -19.866684$$
$$\Sigma\, q_i w_i = 0$$

For this model, multiplying out all four columns verifies the left equation exactly. A matching right vector is $v_i = q_i w_i / N$, where $N = \Sigma q_i w_i^2$. This makes $\Sigma w_i v_i = 1$. Choosing a scale for w changes the numerical overlap but not the hot/cold amplitude ratio.

▼ **Why does a left eigenvector measure an independently decaying component?**

Differentiate its average along a trajectory: $d(w^{\mathsf{T}}p)/dt = w^{\mathsf{T}}p' = w^{\mathsf{T}}Mp = \lambda w^{\mathsf{T}}p$. The first equality uses that w is fixed while the system evolves. This is the scalar equation $y' = \lambda y$ again, so $w^{\mathsf{T}}p(t) = e^{\lambda t}w^{\mathsf{T}}p(0)$. Also $w^{\mathsf{T}}M\pi_b = 0 = \lambda w^{\mathsf{T}}\pi_b$; since λ is nonzero, its equilibrium average is zero. This explains both the exponential behavior and why b can be calculated directly from the initial probabilities.

Make sure it is the slow mode

Being an eigenvector is not enough. There are three decaying modes. The characteristic polynomial, whose roots are eigenvalues, factors as $x(x-\lambda_2)$ times a quadratic. Exact opposite signs at -337 and -336 locate one quadratic root; opposite signs at -60 and -59 locate the other. A quadratic has only two roots, so they are accounted for. Both decay faster than λ_2, which lies between -20 and -19.

▼ **Why do opposite signs locate a root?**

A polynomial is continuous. To go from negative to positive it must pass through zero. The intervals are disjoint, so these are distinct roots. The new checker reconstructs all polynomial coefficients from determinants rather than trusting decimal eigenvalues.

STEP 10 · THE CENTRAL CALCULATION

Average the mode values in each preparation

The slow overlap is $b(\beta) = \Sigma w_i \pi_i(\beta)$. Substitute the weights from Step 3 and values from Step 9. This is the weighted-average calculation from Step 2.

$$b(\beta) = [q_1^{\beta} - 0.759 q_2^{\beta} - 0.6119625 q_3^{\beta} - 0.004 q_4^{\beta}] / \Sigma q_i^{\beta}$$

$$-0.093755177882 < b(0.001) < -0.093755177881$$

$$-0.093857287853 < b(0.01) < -0.093857287852$$

Both are negative, but the hotter overlap has the **smaller absolute value**. It contains less of the pattern that survives longest.

Explore a preparation

Inverse temperature β: 0.001

b = -0.093755177882; |b| = 0.093755177882. T/Tbath = 1000.000.

State	Probability π_i	Mode w_i	Contribution $w_i \pi_i$
1	0.250431646	1	0.250431646

State	Probability π_i	Mode w_i	Contribution $w_i\pi_i$
2	0.250430391	-0.759	-0.190076667
3	0.250202283	-0.6119625	-0.153114415
4	0.248935680	-0.004	-0.000995743

$\beta = 0$ is the infinite-temperature limit, included for exploration. The actual witnesses have positive β. At $\beta = 1$, numerical roundoff may leave a tiny residual near the exact zero.

▼ **How can fractional powers be checked exactly?**

To enclose $q^{1/1000}$, choose positive rational numbers l and u and verify $l^{1000} \le q \le u^{1000}$ with integer arithmetic after clearing denominators. Since positive powers increase, the desired root lies between l and u. The checker finds such bounds by bisection and then carries interval bounds through the numerator and denominator.

STEP 11 · DISTANCE REVERSAL

Connect a smaller overlap to actual relaxation

Define a nonnegative distance-like quantity by squaring each probability error, dividing by q_i, and adding:

$$D_q(p) = \Sigma\ (p_i - \pi_{b,i})^2/q_i.$$

It is zero only at equilibrium. The exact initial values satisfy $D_q(\text{cold}) < 24 < D_q(\text{hot})$. Thus the hotter preparation really starts farther away according to this measure.

At long times, the faster patterns disappear relative to the slowest one. In more precise notation, $p(t) - \pi_b = b(\beta)v\, e^{\lambda_2 t}$ + faster-decaying terms. Insert this into the squared distance. The leading factor shared by both preparations cancels in their ratio:

$$D_q(\text{hot}, t) / D_q(\text{cold}, t) \to [b(\text{hot})/b(\text{cold})]^2 < 1.$$

A limit strictly below one means that after some finite time the ratio stays below one. Initially it is above one. Continuous evolution therefore has at least one crossing in between. This establishes existence, not the earliest crossing time or uniqueness of the crossing.

> **Formal support:** The Lean development proves the evolution equation, uniqueness of its solution, the limiting ordering, and the initial ordering for this rational witness. Its distance is D_q. It does not certify a numerical crossing time or the KL argument below.

▼ **What happens to KL distance?**

Relative entropy, or KL divergence, is $D_{KL}(p\|\pi) = \Sigma p_i \ln(p_i/\pi_i)$. Near equilibrium, the Taylor expansion of ln gives $D_{KL} \approx \frac{1}{2}\Sigma(p_i-\pi_i)^2/\pi_i$. Because $\pi_i = q_i/Q$, this is $QD_q/2$. Here $\text{Var}(E) = \Sigma\pi_i(E_i-U)^2$ is the variance, with mean energy $U = \Sigma\pi_i E_i$. It is positive when distinct energy levels all have positive probability. The same strict late-time ratio follows analytically. For Gibbs preparations, differentiation gives $dD_{KL}/d\beta = (\beta-1)\text{Var}(E)$, negative on the cooling interval $0 < \beta < 1$. So the hotter one also starts farther away in KL. This argument is separate from the Lean distance theorem.

Derive the initial KL ordering instead of memorizing it

1. Let $U(\beta) = \Sigma\pi_i(\beta)E_i$. Differentiating the normalizing sum gives $Z'/Z = -U$.
2. The quotient rule gives $\pi_i' = \pi_i(U-E_i)$. Hence $U' = U^2-\Sigma\pi_i E_i^2 = -\text{Var}(E)$.
3. Because the bath inverse temperature is one, $\ln(\pi_i(\beta)/\pi_i(1)) = (1-\beta)E_i + \ln Z(1)-\ln Z(\beta)$.
4. Average that expression: $D_{KL} = (1-\beta)U + \ln Z(1)-\ln Z(\beta)$.
5. Differentiate. The $-U$ and $+U$ terms cancel, leaving $D_{KL}' = (1-\beta)U' = (\beta-1)\text{Var}(E)$.

For $0 < \beta < 1$ this is strictly negative, so a hotter preparation (smaller β) starts with larger KL distance. This conclusion uses distinct energies and strictly positive Gibbs probabilities.

STEP 12 · INTERPRET THE NUMBERS

Amplitude, distance, and elapsed time are different

Quantity	Definition	Rational witness
Amplitude advantage	$1 - \lvert b_h \rvert / \lvert b_c \rvert$	about 0.10879%
Late-time squared-distance advantage	$1 - (\lvert b_h \rvert / \lvert b_c \rvert)^2$	about 0.21747%
Threshold-time saving	$1 - t_h(\delta)/t_c(\delta)$	Requires a threshold δ and time calculation

If the amplitude ratio approached 3/4, its squared-distance ratio would approach 9/16. That would mean 25% amplitude advantage and 43.75% leading squared-distance advantage, not 25% time saving.

▼ **First-calculus explanation of the time issue**

Suppose $D(t) \approx Cb^2e^{-2\gamma t}$. Solving $D(t) = \delta$ gives $t \approx \ln(Cb^2/\delta)/(2\gamma)$. The cold-minus-hot time difference approaches $\ln(\lvert b_c \rvert / \lvert b_h \rvert)/\gamma$. It is an additive shift. The percentage of total time depends on δ and tends to zero as the threshold tends to zero in this approximation.

Rescaling every rate by a factor c runs the same probability path c times faster. It changes elapsed times, but not the amplitude ratio.

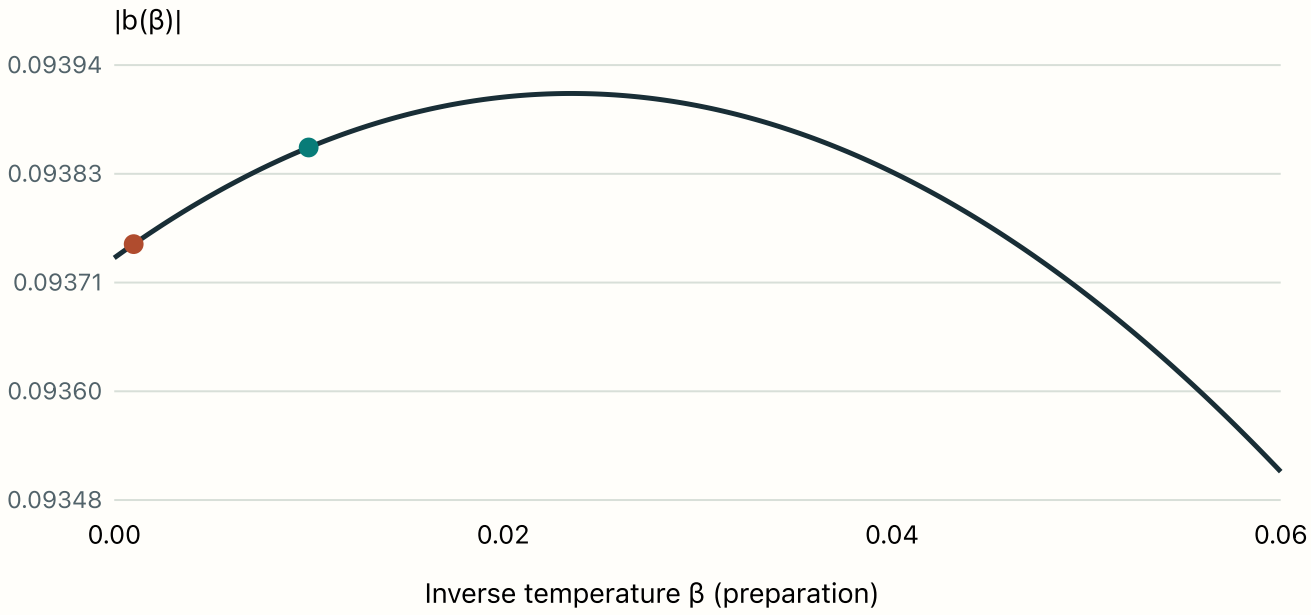
STEP 13 · A MECHANISM YOU CAN SEE

The fourth state dilutes a negative average

The mode values are roughly (1, −0.759, −0.612, −0.004). The highest-energy state contributes almost zero to the average. At very high temperature it nevertheless holds almost one quarter of the probability. That probability dilutes the contribution from the other states.

Cool slightly: the fourth state loses probability rapidly. The negative middle-state contributions can become more prominent, making b more negative. Cool farther toward the bath: the ground state's positive contribution increases and the average returns to zero. This competition allows a small nonmonotonic part of the overlap curve.

A close look near infinite temperature



This curve evaluates the rational witness numerically. It is not a relaxation trajectory: the horizontal axis changes the initial preparation, not elapsed time.

The controlled family varies the fourth mode value z from −0.03 to −0.004, with compensating changes elsewhere to keep a valid slow mode. The existence threshold is approximately $z^* = -0.0179401112$. The fixed hot/cold pair needs $z > -0.014800248$, so it can miss an effect that another pair would detect.

▼ **Why is an endpoint slope enough in this special family?**

Usually it is not. The proof writes $J(\beta) = e^{\beta E_4} Z(\beta)^2 b'(\beta)$, expands it into exponential terms, and shows J is strictly increasing with $J(1) > 0$. Thus a negative $J(0)$ gives exactly one interior zero; a nonnegative $J(0)$ gives none. That extra monotonicity proof rules out hidden turns. The rate and spectral checks must also hold throughout the family, not just at the plotted points.

▼ **Slow route: prove the single-turn test using first-year calculus**

1. Separate the numerator from its positive denominator

Use energies $0 < E_2 < E_3 < E_4$, a bath at $\beta = 1$, and mode values $w = (1, -p, -r, z)$, where $p > r > -z > 0$. Here p and r are positive numbers, while z is negative. This proof applies when $E_4 > E_2+E_3$ and $(1+r)E_3 > (p-r)(E_3-E_2)$. These are extra hypotheses of this particular mechanism.

$$N(\beta) = 1 - p\,e^{-\beta E_2} - r\,e^{-\beta E_3} + z\,e^{-\beta E_4},$$
$$Z(\beta) = 1 + e^{-\beta E_2} + e^{-\beta E_3} + e^{-\beta E_4}, \quad b = N/Z.$$

Differentiating N gives three positive terms: $pE_2e^{-\beta E_2}$, $rE_3e^{-\beta E_3}$, and $-zE_4e^{-\beta E_4}$. Thus N increases strictly. Bath orthogonality says $N(1) = 0$. Consequently b is negative before the bath and positive after it. There is no other zero.

2. Turn the quotient rule into a sign problem

Set $D = N'Z - NZ'$. Since Z is positive, the quotient rule gives $b' = D/Z^2$; D and b′ have exactly the same sign. Expanding the two products pairs states i and j. Their contribution is $(w_i - w_j)(E_j - E_i)e^{-\beta(E_i+E_j)}$. Define six positive constants:

$$c_{1j} = (1-w_j)E_j \quad \text{for } j = 2,3,4;$$
$$c_{ij} = (w_j-w_i)(E_j-E_i) \quad \text{for } 2 \le i < j \le 4.$$

The second group is positive because $-p < -r < z$. The extra energy inequality above is precisely $c_{13} > c_{23}$. Multiply D by the positive factor $e^{\beta E_4}$ and regroup:

$$J(\beta) = c_{12}e^{\beta(E_4-E_2)}$$
$$+ e^{\beta(E_4-E_2-E_3)}(c_{13}e^{\beta E_2} - c_{23})$$
$$+ c_{14} - c_{24}e^{-\beta E_2} - c_{34}e^{-\beta E_3}.$$

For $\beta \ge 0$, the first term increases. Both factors in the second term are positive and increasing. The constant stays constant. Each of the last two negative terms increases toward zero. Therefore J is strictly

increasing. Also $J(1) > 0$, because $N(1) = 0$ and $N'(1) > 0$.

3. Check the endpoint, including equality

$$J(0) = D_0 = (1-p-r+z)(E_2+E_3+E_4) + 4(pE_2+rE_3-zE_4).$$

If $D_0 < 0$, continuity gives a zero between 0 and 1, and strict increase makes that zero unique. Before it, $b' < 0$. Since $b < 0$, the magnitude $|b|$ increases there. Any two distinct positive β values in that interval give finite hot and cold preparations with the anomalous ordering. After the zero, $|b|$ decreases back to zero at the bath.

If $D_0 = 0$, strict increase gives $J(\beta) > 0$ at every $\beta > 0$. If $D_0 > 0$, the same conclusion holds. Both cases exclude a cooling turn. On the heating side $\beta > 1$, both b and b' are positive, giving normal heating order.

4. Keep the physical generator attached to the curve

A desired curve is not enough: it must come from the slowest nonconstant mode of an allowed rate matrix. In the controlled family, $r = r_0+\eta z$, where $r_0 = 48959/80000$ and $\eta = 1/160$. This adjustment keeps the bath mean zero. Two rates are solved from the eigenvector equation while four stay fixed. Exact bounds show that every rate remains positive and every original triplet test fails throughout $-0.03 \le z \le -0.004$.

The remaining spectral check uses a baseline matrix. After converting detailed balance into a symmetric matrix, ordinary elimination rewrites its quadratic expression as a sum of four nonzero constants times squares. For the baseline shifted by 50, two constants are negative. An invertible change of variables preserves the maximum number of independent negative directions, so exactly two baseline eigenvalues lie below -50. Increasing the two varying rates only adds nonnegative squared differences to the decay expression. Those two eigenvalues therefore stay below -50. The constructed eigenvalue lies between -20 and -19; the fourth is zero. Thus the constructed mode really is the unique slow nonconstant mode.

Substitution gives $D_0 = U-Vz$ with $V > 0$. Hence the exact onset is $z^* = U/V$. Its decimal enclosure comes from rational upper and lower bounds for the logarithms of the bath weights. The separate fixed-pair threshold solves $b_h-b_c = 0$; it is different because a particular pair can miss a short interval of anomalous ordering.

5. Why the effect initially grows quadratically

Write $\delta = z-z^*$ and $d^* = \partial D/\partial\beta$ at $(0,z^*)$. The increasing-J proof gives $d^* > 0$. Near the onset, the leading terms are $D \approx -V\delta+d^*\beta$. Solving $D = 0$ gives $\beta_m \approx V\delta/d^*$. Because $Z(0) = 4$, integrate $b' \approx (-V\delta+d^*\beta)/16$

from 0 to β_m. The change in b is $-V^2\delta^2/(32d^*)$, so its negative magnitude grows by $V^2\delta^2/(32d^*)$. Smoothness supplies errors of order δ^2 in the peak location and δ^3 in its height. This approximation explains the onset; the exact sign proof above establishes existence.

Inspect the full derivation, continuous arithmetic certificate, and step-by-step review record.

STEP 14 · THE 25% BOUND

Separate the difficult mode theorem from easy averaging

The difficult part of the theorem proves that the triplet premise restricts the slow mode. With a suitable sign choice:

$$w_1 \geq w_2, \quad w_1+w_2 \geq 2w_3, \quad w_4 \leq 0,$$
$$\text{and } b(\beta) < 0 \text{ for } 0 < \beta < 1.$$

These conditions are conclusions about a *simple* slow mode; simple means its eigenvalue occurs only once. They are not assumptions one is free to add to the conjecture. Their proof uses the full triplet constraints, a change of coordinates, and an elimination argument. The Lean library checks that structural step. The rest can be followed using weighted averages.

1. Let P(β) be the total probability in states 1, 2, 3. Each has lower energy than state 4, so $3/4 < P < 1$ for finite $\beta > 0$. Cooling increases P.
2. Let m(β) be the conditional average of w within these three states: divide their weighted sum by their total probability.
3. The first two mode inequalities imply that m cannot decrease as β increases. The expansion in the box below proves this.
4. The full average is $b = Pm + (1-P)w_4$.

▼ **Check the conditional-average derivative**

Write (w_1,w_2,w_3) as a constant vector plus $a(1,-1,0)+c(1,1,-2)$, where $a = (w_1-w_2)/2 \geq 0$ and $c = (w_1+w_2-2w_3)/6 \geq 0$. Put $x = e^{-\beta E_2}$, $y = e^{-\beta E_3}$. The average of the first pattern is $(1-x)/(1+x+y)$. Its derivative has numerator $2E_2x + E_3y - (E_3-E_2)xy > 0$, since $x \leq 1$. The second average is $1-3y/(1+x+y)$, also

increasing because the third state's conditional probability decreases. Nonnegative combinations stay nondecreasing.

▼ **Prove the two claims about P**

Each of the first three Boltzmann weights is strictly larger than the fourth when $\beta > 0$ and the energies are ordered. Therefore the fourth probability is less than 1/4, so $P = 1-\pi_4 > 3/4$. It is also less than one, because π_4 is positive. For monotonicity, differentiate π_4: $\pi_4' = \pi_4(U-E_4)$. The mean energy U is strictly below the largest energy E_4, so $\pi_4' < 0$ and $P' > 0$. All inequalities are strict for the finite admissible parameters.

▼ **Slow route: weighted coordinates explain the sign of the slow overlap**

1. Replace a difficult average with simple comparisons

Here q is the bath probability vector: each $q_i > 0$ and their sum is one. Define $Q_k = q_1+\cdots+q_k$. A prefix contrast compares the first k states with the next state:

$$h^{(k)} = (1, \ldots, 1, -Q_k/q_{k+1}, 0, \ldots, 0).$$

There are k initial ones. The bath-weighted average is $Q_k - q_{k+1}(Q_k/q_{k+1}) = 0$. For four states, these are $(1,-q_1/q_2,0,0)$, $(1,1,-Q_2/q_3,0)$, and $(1,1,1,-Q_3/q_4)$.

The weighted dot product of vectors f and g is $\Sigma q_i f_i g_i$. It is the ordinary dot product after multiplying coordinate i by $\sqrt{q_i}$. Any two different prefix contrasts have weighted dot product zero: the longer contrast is constant on the support of the shorter, whose weighted sum is zero. Each squared length is $\eta_k = Q_k Q_{k+1}/q_{k+1}$. Dividing $h^{(k)}$ by $\sqrt{\eta_k}$ produces a unit vector $e^{(k)}$.

These n−1 nonzero perpendicular vectors give coordinates for every vector with bath mean zero. This is just the familiar idea of perpendicular x and y axes, applied to a space with n−1 directions.

2. Translate the rate inequalities

Detailed balance makes $k_{ij} = M_{ij}/q_i$ symmetric. The triplet quantity G_{ijk} is $q_j(k_{ik}-k_{jk})$. Thus $G \le 0$ says $k_{ik} \le k_{jk}$ whenever $i < j < k$. Write the generator acting on functions as $L = M^T$. In the unit contrast coordinates its matrix is $A_{kl} = \Sigma q_i e_i^{(k)} (Le^{(l)})_i$. Detailed balance makes A symmetric.

For $k < l$, direct multiplication gives the following off-diagonal entry:

$$\sqrt{(\eta_k \eta_l)}\, A_{kl} = Q_{l+1} \sum_{i \le k} q_i (k_{k+1,l+1} - k_{i,l+1}) + \sum_{j \ge l+2} q_j \sum_{i \le k} q_i (k_{k+1,j} - k_{ij}) \ge 0.$$

Every displayed difference is nonnegative by $G \le 0$. To reproduce the multiplication, first observe that for $i \le k+1 \le l$, $(Lh^{(l)})_i = -Q_{l+1} k_{i,l+1} - \sum_{j \ge l+2} q_j k_{ij}$. Taking the shorter contrast subtracts the value at $k+1$ from the weighted prefix values.

3. Why nonnegative coordinates stay nonnegative

The off-diagonal entries of A are nonnegative; its diagonal entries can be negative. Choose c large enough that every entry of $A+cI$ is nonnegative. The exponential series $I+t(A+cI)+t^2(A+cI)^2/2!+\cdots$ then has nonnegative entries and diagonal entries at least one. Multiplying by e^{-ct} shows e^{tA} has nonnegative entries and positive diagonal entries for every finite $t \ge 0$. This is a direct reason that componentwise positive vectors remain positive.

The largest-eigenvalue vector of a symmetric A can be chosen nonnegative. One elementary explanation is to maximize $x^{\mathsf{T}}Ax$ over unit vectors. Replacing each coordinate by its absolute value cannot lower that expression, since its off-diagonal coefficients are nonnegative. At a maximum, the usual constrained derivative calculation gives $Ax = \lambda x$. Maximizing selects the largest eigenvalue. No claim that every eigenvector has one sign is being made.

4. Compute the thermal contrast coordinates

Let $g_k(\beta) = \sum e_i^{(k)} \pi_i(\beta)$. Apart from the positive factor $Z\sqrt{\eta_k}$, its numerator is

$$N_k(\beta) = e^{-\beta E_{k+1}} \sum_{i \le k} \left[e^{\beta (E_{k+1} - E_i)} - e^{\beta_b (E_{k+1} - E_i)} \right].$$

All energy gaps in brackets are positive. Thus all coordinates g_k are negative on cooling, zero at the bath, and positive on heating. A nonzero nonnegative slow eigenvector has strictly negative dot product with g on cooling and strictly positive dot product on heating. That proves the overlap sign used in Step 14.

On heating the coordinates also increase strictly. Write $Z = N_k + T_k$, where $T_k = (1 + Q_k/q_{k+1}) e^{-\beta E_{k+1}} + \sum_{j \ge k+2} e^{-\beta E_j}$. Both parts are positive. Their logarithmic derivatives satisfy $N_k'/N_k > -E_{k+1}$ and $T_k'/T_k \le -E_{k+1}$. The first follows from the increasing positive bracket above; the second is a weighted average of negative energies. The quotient rule now gives $(N_k/Z)' > 0$.

5. What follows when the slow eigenvalue repeats?

A single overlap is then insufficient. Instead use the projection Π onto the entire slow eigenspace. Symmetric diagonalization gives $\Pi = \lim_{t\to\infty} e^{-\lambda_2 t} e^{tA}$. The matrices inside the limit are nonnegative, so Π is nonnegative too. It is nonzero and has a positive diagonal entry. A vector whose entries are all strictly positive, or all strictly negative, therefore cannot project to zero. This excludes strong effects even with a repeated slow eigenvalue.

For heating, farther preparations have strictly larger positive g coordinates. Applying e^{tA} preserves that order. The squared weighted distance is the sum of the squares of these coordinates, so it stays strictly larger at every finite time. The leading KL coefficient is proportional to $\|\Pi g\|^2$ and also increases on heating. This proves the stated all-time squared-distance and asymptotic KL conclusions. It does not establish all-time KL or total-variation order.

Read the general proof and symbolic identity checks. The code checks dimensions 2 through 7; the index calculation above is the argument for arbitrary finite n.

▼ **Slow route: derive the difficult first-three mode restriction**

This route builds on the weighted-coordinate explanation immediately above. It uses algebra, inequalities, and the largest-eigenvalue idea. The new matrix-elimination step is explained along the way. Its hypotheses are four states, strictly ordered energies, positive reversible rates, every F and G nonpositive, and a simple slow eigenvalue.

1. Isolate the one inequality that needs proof

Use the unnormalized contrasts as columns of H. Write the slow mode $w = Hx$, where $x = (x_1, x_2, x_3)$ has nonnegative entries. The matrix in these coordinates is $B = \mathrm{diag}(\eta)^{-1} H^{T} \mathrm{diag}(q) L H$. Multiplying coordinate k by $\sqrt{\eta_k}$ turns B into the symmetric matrix A above, so it has the same eigenvalues.

$$
\begin{aligned}
&w_1 - w_2 = (Q_2/q_2)x_1 \geq 0, \quad w_4 = -(Q_3/q_4)x_3 \leq 0,\\
&w_1 + w_2 - 2w_3 = (2Q_3/q_3)(x_2 - s x_1),\\
&s = (q_1 - q_2)q_3/(2q_2Q_3) > 0.
\end{aligned}
$$

Only $x_2 \geq s x_1$ remains. If $x_1 = 0$, it already holds.

2. Subtract a common mixing process

The reduced conductances $k_{ij} = a_{ij}/q_i$ have their minimum in the first row, because $G \le 0$ orders each column. Let k^* be the smallest of k_{12}, k_{13}, k_{14}. Subtract the reset generator $k^*(q\mathbf{1}^T - Q_4 I)$. Its off-diagonal rate into state i is $k^* q_i$. All remaining rates are nonnegative, and at least one of rates 12, 13, 14 is now zero.

This operation leaves F and G unchanged: substituting $a_{ij} - k^* q_i$ makes the extra terms cancel. On every bath-mean-zero mode it merely adds $k^* Q_4$ to the eigenvalue. Eigenvectors, ordering and simplicity are unchanged. The remaining generator may disconnect; the proof allows that.

Its eigenvalues are at most zero. To see why, detailed balance gives $-\langle f, Lf\rangle_q = \Sigma_{i<j} q_i q_j k_{ij} (f_i - f_j)^2 \ge 0$ for the remaining conductances. For an eigenvector the left side is $-\lambda\|f\|^2$. Thus $\lambda \le 0$, even when some edges vanish.

3. Eliminate the third unknown safely

All following B entries refer to the remaining generator. Put $\delta = \lambda - B_{33}$, where λ is its largest contrast eigenvalue. We need $\delta > 0$ before dividing. Symmetry implies $\lambda \ge B_{33}$: test the quadratic expression on the third unit vector. If equality held, the third diagonal entry of $\lambda I - A$ would be zero. This matrix has nonnegative quadratic expression; any nonzero off-diagonal entry in that row would make the expression negative for a suitable small perturbation. Its whole third row must therefore vanish. The third unit vector would be a largest-eigenvalue vector. Simplicity would force $x_1 = 0$, contrary to the case under consideration. Hence $\delta > 0$.

The third equation gives $x_3 = (B_{31}x_1 + B_{32}x_2)/\delta$. Substitute it into the first two equations. The resulting two-by-two matrix is $E_{ij} = B_{ij} + B_{i3}B_{3j}/\delta$, for $i,j = 1,2$.

Why is λ still its simple largest eigenvalue? In symmetric coordinates, complete the square in the third variable of the nonnegative quadratic expression for $\lambda I - A$. The remaining two-variable expression is exactly the symmetric version of $\lambda I - E$. It is nonnegative. Completing the square is an invertible change of variables, and the eliminated square has positive coefficient δ, so the dimension of the zero space remains one. This gives both largest-eigenvalue ordering and simplicity.

4. Turn the eigenvector ratio into a polynomial sign

$$
\begin{aligned}
f(s) &= E_{21} + s(E_{22} - E_{11}) - s^2 E_{12}, \\
\delta f(s) &= \delta C + D^*, \\
C &= B_{21} + s(B_{22} - B_{11}) - s^2 B_{12}, \\
D^* &= (B_{23} - sB_{13})(B_{31} + sB_{32}).
\end{aligned}
$$

If the off-diagonal entries of E are positive, its eigenvector ratio x_2/x_1 is the unique positive zero of f. The quadratic begins positive at zero and opens downward. Therefore $f(s) \geq 0$ implies $s \leq x_2/x_1$. If one off-diagonal entry vanishes, symmetry after positive rescaling makes both vanish. Simplicity and $x_1 > 0$ then give $x_2 = 0$ and $E_{11} > E_{22}$, so $f(s) < 0$. Thus showing $f(s) \geq 0$ also excludes this exceptional case.

The G inequalities give $B_{31}, B_{32} \geq 0$ and

$$B_{23} - sB_{13} = [Q_4/(2q_1q_2Q_3)](2a_{34}q_1q_2 - a_{14}q_2q_3 - a_{24}q_1q_3) \geq 0.$$

For the last sign, $G_{134} \leq 0$ and $G_{234} \leq 0$ bound each of the two subtracted terms by $a_{34}q_1q_2$. Hence $D^* \geq 0$. If $C \geq 0$, we are done. If $C < 0$, use $\lambda \leq 0$ to get $\delta C + D^* \geq -B_{33}C + D^*$. The unknown eigenvalue has disappeared.

5. Check every zero-rate boundary

Call the six remaining downward rates $(a,b,c,d,e,f) = (a_{12}, a_{13}, a_{14}, a_{23}, a_{24}, a_{34})$. Here the rate f is unrelated to the polynomial f(s) above. Direct expansion gives

$$-B_{33}C + D^* = [Q_2Q_4/(4q_1q_2^2Q_3^2)]T,$$
$$T = (c+e+f)N + q_4[2ceq_3(q_1 - q_2) + fJ],$$
$$N = -2q_1q_2q_3F_{123} \geq 0,$$
$$J = eq_1(2q_2 + q_3) - cq_2(2q_1 + q_3).$$

The prefactor is positive. At least one of a,b,c is zero; this is exactly why we subtracted the reset process.

1. **c = 0:** $J = eq_1(2q_2 + q_3) \geq 0$. Every term in T is nonnegative.
2. **a = 0:** $F_{124} \leq 0$ gives $eq_1(2q_2 + q_4) - cq_2(2q_1 + q_4) \geq 0$. Multiply this by $(2q_1 + q_3)/(2q_1 + q_4)$, then add $2eq_1(q_1 - q_2)(q_3 - q_4)/(2q_1 + q_4)$. The result is exactly J. Both pieces are nonnegative, including when c or e is zero. Thus $T \geq 0$.
3. **b = 0:** $N = 2aq_2q_3(q_1 - q_2) + dq_1q_3(2q_2 + q_3)$. If $c \leq e$, write $J = (e-c)q_1(2q_2 + q_3) + cq_3(q_1 - q_2) \geq 0$. If $c > e$, use the identity $N + q_4J = 2q_1q_2(q_3 - q_4)(c - e) + dq_1q_3(2q_2 + q_3) - 2q_1q_2q_3F_{124} \geq 0$. Now $T = (c+e)N + f(N + q_4J) + 2ceq_3q_4(q_1 - q_2) \geq 0$.

These cases can overlap, but they cover every possibility. We have proved $\delta f(s) \geq 0$; $\delta > 0$ then gives $x_2 \geq sx_1$. This completes the missing mode inequality, including disconnected remaining generators and

equality in the triplet tests. The weighted-average argument in Step 14 now supplies the strict 25% upper bound. Showing that systems can approach this bound is a separate construction in Step 15.

The long algebraic identities can be checked in the exact symbolic certificate; the formal developments begin in ModeShape.lean. The review record states exactly which results have been checked.

For a hot and cold pair, let $r = P_c/P_h$. Then $1 < r < 4/3$. Subtract the two averages:

$$b_c - r\,b_h = P_c(m_c - m_h) + (1-r)w_4 \geq 0.$$

Both terms are nonnegative. Since both b's are negative, this says $|b_c| \leq r|b_h| < (4/3)|b_h|$. Therefore $|b_h|/|b_c| > 3/4$, and the amplitude advantage is strictly less than 1/4.

STEP 15 · SHARPNESS

“Sharp” means arbitrarily close is possible

A bound can be true but wasteful. To show that 25% is sharp, build valid systems approaching it. The formal family uses a small positive ε:

$$q = (1,\ 1-\varepsilon/3,\ 3/5,\ \varepsilon^2)$$
$$w = (1,\ -4/5,\ -1/3-4\varepsilon/9+5\varepsilon^3/3,\ -\varepsilon).$$

Choose $\varepsilon = e^{-L^2}$, $\beta_h = L^{-3}$, $\beta_c = L^{-1}$, with L large and finite. The highest energy is $2L^2$. Its hot weight is $e^{-2/L}$, approaching one, while its cold weight is e^{-2L}, approaching zero. The first three weights approach one in both preparations.

$$1 - 4/5 - 1/3 = -2/15$$

$$b_h \to (-2/15)/4 = -1/30$$

$$b_c \to (-2/15)/3 = -2/45$$

$$1 - |b_h|/|b_c| \to 1 - 3/4 = 1/4.$$

This limiting arithmetic alone is not enough: the family must have positive rates, the required triplet signs, and a simple slow mode for sufficiently small positive ε. Those conditions are part of the formal proof. Some triplet expressions are exactly zero, which is permitted by the original strict tests. A different, separately certified family makes all the inequalities strict and retains the same limit.

▼ **Slow route: prove that valid systems really approach 25%**

1. Keep two different constructions separate

Both constructions use the q and w printed above, but their rates differ. The simpler family is formalized in Lean and proves that 25% is the supremum. A second family proves the stronger claim that the slow mode can stay uniformly separated from faster modes. A small perturbation of that second family makes all eight F and G inequalities strict. The stronger gap and perturbation claims are checked analytically and by exact arithmetic, outside the Lean theorem.

2. A continuity argument gives actual positive parameters

For the simpler family, set rate $a_{12} = 1$ and define the other rates by the rational functions in the formal sharpness appendix. A rational function is a polynomial divided by a polynomial. If its denominator is nonzero at $\varepsilon = 0$, its value changes continuously near zero. If its value there is positive, it remains positive on some interval around zero.

The proof needs thirteen such functions to be positive simultaneously. They control the three polynomial denominators, five varying rates, three F signs and two G signs. Their values at zero, in the appendix's order, are

$$68,\ 12,\ 21,\ 3/2,\ 20/17,\ 15/7,\ 20/17,\ 12/17,$$
$$9/14,\ 117/70,\ 66/425,\ 276/595,\ 1/17.$$

All are positive. Each has some positive interval where its sign persists. The smallest of finitely many positive interval widths is still positive, so all thirteen conditions hold together for sufficiently small $\varepsilon > 0$.

This is an existence proof for an interval; it does not claim that substituting $\varepsilon = 0$ gives a valid bath.

Two triplet expressions, F_{124} and G_{234}, are exactly zero. Another, G_{124}, tends to zero. Its sign is established by factoring out ε: $-G_{124}$ is ε times the thirteenth positive function. $G_{134} < 0$ follows from $G_{234} = 0$ and $G_{124} < 0$. This accounts for all eight F and G expressions, including the ones continuity alone would not decide.

For small positive ε, the bath weights are positive and strictly decreasing. Exact substitution checks conservation, detailed balance, and $M^{T}w = -3w$. The five rate limits are positive as well. No numerical eigenvalue calculation is needed.

3. Verify that −3 is the simple slow eigenvalue

The three contrast coordinates of w are positive. The contrast matrix B has nonnegative off-diagonal entries, and the two entries connecting its first coordinate to each of the other two are strictly positive. Here is a useful direct algebraic test for simplicity.

Let $A = \mathrm{diag}(\eta)(-3I-B)$. It is symmetric, $Ax = 0$ for the positive coordinate vector x, and its off-diagonal entries are nonpositive. Expanding the squares gives, for every y,

$$y^{T}Ay = \Sigma_{i<j}\ (-A_{ij})x_i x_j (y_i/x_i - y_j/x_j)^2 \geq 0.$$

Thus −3 is at least as large as every contrast eigenvalue. Equality in this sum forces $y_1/x_1 = y_2/x_2$ and $y_1/x_1 = y_3/x_3$, because the two connecting coefficients are positive. Hence every zero vector of A is a multiple of x: the eigenvalue is simple. All nonstationary modes lie in this contrast space, so this is the desired slow-mode statement for the full generator.

4. Join admissibility to the thermal limit

The parameter $\varepsilon = e^{-L^2}$ is positive for each finite L and tends to zero as L grows. It eventually enters the interval where every rate and sign check holds. At the same time, $L > 1$ gives $0 < L^{-3} < L^{-1} < 1$, so both preparations are finite and on the cooling side.

The overlap limits above are nonzero and have a strict magnitude difference of 1/90. Therefore sufficiently late members have a strict ordinary cooling reversal. Their advantages approach 1/4. This proves that any proposed upper bound below 1/4 eventually fails, while Step 14 proves every actual advantage remains below 1/4. Those are the two parts of a supremum proof.

5. Why the stronger family has a uniform gap

The second construction chooses two vectors U that are perpendicular, in the bath-weighted dot product, to both the constant vector and w. They span the two faster directions. A symmetric two-by-two matrix B is chosen using the linear equations $F_{124} = G_{234} = 0$. Here B denotes this new matrix, not the contrast matrix in the previous paragraphs.

$$M = -\mathrm{diag}(q)\, U(B/\varepsilon^2)U^{\mathsf{T}} + 3(q1^{\mathsf{T}} - I).$$

The first term leaves the constant and slow directions alone; the reset term gives w eigenvalue −3. In the faster space the extra decay rates are the two positive eigenvalues ν_1,ν_2 of BJ/ε^2, where $J = U^{\mathsf{T}}\mathrm{diag}(q)U$. Exact inequalities show B is positive definite: $B_{11} > 0$ and $\det B > 0$. The columns of U are independent, so J is positive definite too. Their product has positive real eigenvalues because it is similar to a symmetric positive definite matrix.

For any positive numbers ν_1,ν_2, their product divided by their sum is strictly smaller than either one. The certificate proves $\det(BJ/\varepsilon^2)/\mathrm{trace}(BJ/\varepsilon^2) > 1/3$. Thus both faster eigenvalues lie below −3−1/3. The same certificate verifies all twelve positive rates and every triplet sign for $0 < \varepsilon \le 1/30$, using thirty exact polynomial sign certificates.

6. Make the triplet inequalities strict without losing the limit

Add $\varepsilon^4 H$, where H is a reversible generator with all eight triplet expressions strictly negative and reduced conductances at most 100. Its largest possible effect on a unit-vector quadratic expression is at most $100\varepsilon^4$. The old slow-to-fast gap therefore shrinks by at most $200\varepsilon^4$, remaining above 1/4.

Projecting the perturbed slow vector onto the old faster space bounds that component by $100\varepsilon^4/(1/3-100\varepsilon^4)$. After matching sign and weighted length, the total weighted vector change is less than $500\varepsilon^4$. Each bath probability is at least $\varepsilon^2/3$, so each ordinary coordinate changes by less than $1000\varepsilon^3$. An average against any probability distribution changes by no more than that largest coordinate error. The error tends to zero, leaving the overlap limits intact. Linearity of F and G makes every formerly zero constraint strictly negative.

The finite example $\varepsilon = 10^{-50}$, $\beta_h = 1/10000$ and $\beta_c = 1/50$ has a certified amplitude advantage between 22.6171597478% and 22.6171597506% before perturbation, and above 22% afterward. This is a finite mathematical example with very large energy separation.

Inspect the analytic review, 21 exact correspondence checks, and uniform-family certificate.

▼ **A useful restriction when the energy span is limited**

Let $\Delta = E_4 - E_1$ and $d = (\beta_c - \beta_h)\Delta$. Write $p_{4,h}$ for the hottest preparation's probability in the highest state. Step 14 actually proves $|b_c|/|b_h| \le P_c/P_h$, where P is the first-three probability mass. Invert these positive ratios to obtain an advantage at most $1 - P_h/P_c$.

Shift E_1 to zero. The sum Z_3 of the first three unnormalized weights decreases on cooling, while the fourth weight changes by the factor e^{-d}. Consequently $\pi_{4,c}/P_c \ge e^{-d} p_{4,h}/P_h$. Rearranging gives

$$\text{Advantage} \le p_{4,h}(1-e^{-d}) < (1-e^{-d})/4.$$

To achieve an advantage at least $g > 0$, it is therefore necessary that $d > -\ln(1-4g)$ and $p_{4,h} > g$. For 10%, 20% and 24%, the required d values are respectively greater than $\ln(5/3)$, $\ln(5)$, and $\ln(25)$. These conditions only rule out impossible targets; satisfying them does not guarantee an effect.

No finite admissible member attains 25%. The limiting endpoint has singular parameters and is not a valid substitute for a finite witness. Large advantages here require large energy separation.

STEP 16 · A DIFFERENT RESEARCH QUESTION

Finite-time gains in a shared-barrier model

The later physical-model study asks for visible finite-time improvement at more moderate parameter values. It uses energies $(0, g_1, g_1+g_2, g_1+g_2+g_3)$, two basins $\{1,4\}$ and $\{2,3\}$, and three shared barrier excesses. A basin here is a group of configurations with relatively fast internal rearrangement.

$$B_{ij} = \max(E_i, E_j) + h_{ij}$$
$$M_{ij} = \nu_0 \exp[-(B_{ij} - E_j)]$$
$$h_{14} = h_A,\ h_{23} = h_B,\ \text{other } h_{ij} = h_X.$$

B is symmetric between a pair, so $M_{ij}e^{-E_j} = \nu_0 e^{-B_{ij}} = M_{ji}e^{-E_i}$. This verifies detailed balance directly. The attempt frequency ν_0 sets the clock.

The nominal temperatures are about 3.8694 and 9.4721 times the bath temperature. The hotter preparation has less of the slow mismatch between basin populations, despite its larger initial KL distance. The reported continuous certificate establishes:

- At least 10% KL-distance advantage for every $0.4 \le t/\tau \le 0.8$, with cold KL at least 0.001.
- At least 5% time saving for every threshold $0.001 \le \delta \le 0.01$.
- Both conclusions throughout the specified simultaneous ±5% uncertainty box.

Here τ is the slow relaxation time of the perturbed generator itself. The uncertainty coordinates are the three energy gaps, three barrier excesses, and the two **temperature excesses $T/T_b - 1$**. They are not ±5% changes in eight arbitrary matrix entries.

> Two triplets pass in this model. It is not another all-triplet-failure counterexample. Its exact stochastic bath realization is an idealized model; it does not establish experimental cooling, a material's rate law, or a net benefit after preparation costs.

STEP 17 · CONTINUOUS VERIFICATION

Checking many points does not check every point

There are infinitely many points in a box of possible parameters. Even checking all corners can miss an interior failure. Interval arithmetic replaces a number with a lower and upper bound and carries safe bounds through each operation. If a full small box proves the required sign, every point in it passes. If it cannot, split the box and try again.

```
Example: x ∈ [1,2], y ∈ [3,4] ⇒ x+y ∈ [4,6].
But x−x may be enclosed as [−1,1], even though its actual value is 0.
```

The second line shows dependency overestimation. An interval engine may fail to prove a true claim because its enclosure is too wide. An unresolved box is neither a proof nor a physical counterexample. Outward rounding, bounds on truncated series, and a complete subdivision tree all matter.

Why eight proof obligations?

1. Prove the distance advantage throughout its time window.
2. Prove both preparations are above the threshold band early.
3. Prove the cold one is below the band late.
4. Prove cold KL is concave as a function of cold temperature over the needed domain.
5. Prove hot KL is nonincreasing in hot temperature at threshold crossings.
6. Prove the hot one is below the band late.
7. Prove the time comparison at the low-cold/low-hot temperature endpoints.
8. Prove it at the high-cold/low-hot endpoints.

▼ **Follow the temperature reduction with freshman calculus**

At a fixed time, a concave curve lies above the chord connecting its endpoints. Let m be the earlier of the two cold endpoint threshold times. At time m, both endpoints are at or above the threshold, so every interior cold temperature is too. No interior cold temperature arrives earlier than m.

For the hot trajectory, differentiate $D(T,u(T)) = \delta$: $u'(T) = -D_T/D_u$. The certificate establishes $D_T \le 0$ at these roots, and relaxation gives $D_u < 0$. Hence $u' \le 0$: the lowest hot temperature is slowest. Verifying both cold endpoints against this worst hot endpoint therefore covers all preparation temperatures. The remaining six physical parameters still need continuous coverage.

A successful replay must end with no pending cells and no enclosure errors. The replay alone does not prove that every formula in the implementation is correct: this review also needs the derivations, rounding behavior, input correspondence, independent calculations, and negative controls.

STEP 18 · MAP THE REST OF THE PROJECT

Broader results need their own assumptions

The long report contains more than the focused counterexample. This map introduces the questions; it does not replace the detailed proofs. The numbered-claim ledger in Step 20 keeps their review open until their complete arguments have been checked.

Topic	Plain-language meaning	Scope to preserve
G-only constraints	Use only one of the two triplet inequalities to control signs of slow modes.	A weaker premise, so its bounds can differ from the full F-and-G theorem.
Strong effects	A nonbath preparation has exactly zero slow component.	Exact zero is stronger than a small nonzero component. A floating tolerance cannot decide it.
Classification	A finite decision procedure determines which thermal overlap curves can turn or vanish.	Simple slow modes and repeated slow modes require different treatment; asymptotic classification is not all-time classification.
Commensurate energies	All energies are integer multiples of one energy unit.	Changing variables converts exponential expressions into polynomials whose roots can be counted exactly.
Rational-input endpoint decision	For a special premise, one endpoint sign decides ordinary cooling.	Uses algebraic numbers and logarithms, including a stated transcendence theorem; it is not a generic unconstrained classifier.
Repeated slow eigenvalues	More than one independent slow pattern survives at the same rate.	A single scalar overlap is insufficient. The slow-space projection and chosen distance matter.
Inverse feasibility	Start with desired mode values and ask whether any allowed rates realize them.	The eigenvector equation alone is insufficient; the vector must also be slowest and all rate constraints must hold.
Finite-time tests	A faster mode can create a temporary crossing even with normal eventual order.	The complete specified-pair test is for weighted Euclidean distance, not every distance.
Near-bath KL window	Small enough thermal perturbations satisfy a normal finite-time order under the stated G premise.	The sufficient window uses $\lvert\beta-\beta_b\rvert(E_n-E_1) \le 2/\sqrt{5}$; no claim that this constant is optimal.

Topic	Plain-language meaning	Scope to preserve
More than four states	Construct examples approaching amplitude advantage 1–3/n.	The general upper bound 1–2/n leaves a gap for n > 4. Sharpness is established here only for n = 4.
Obstructions	A counterexample breaks an attempted method of proof.	Breaking a proof method does not necessarily refute the target theorem or produce a thermal effect.
Activated intermediates	Add short-lived configurations to examine the physical clock and coarse graining.	The expanded model lies outside the original four-state parameter box; sampled behavior is not a continuous certificate.

Read the reproduction map and comprehensive report for the exact hypotheses. Advanced techniques are introduced where needed in the full audit; familiarity with the glossary does not make those proofs automatic.

▼ **The report's original example: how to check an irrational slow mode**

The long report's primary example and the focused paper's example use the same bath weights, but different rates 23 and 34. The primary example uses $a_{23} = 1799/8$ and $a_{34} = 251/25$. Its slow eigenvalue is a root of a cubic rather than a rational number. Do not mix the two examples' eigenvectors or overlap decimals.

1. Locate a root without trusting a decimal approximation

The exact characteristic polynomial factors into x times a cubic. Substitute the rational endpoints of (−337,−336), (−60,−59), and (−20,−19). The cubic has opposite signs at the ends of each interval. Continuity puts a root in each. A cubic has only three roots counting multiplicity, so these three distinct roots exhaust it and each is simple. The root closest to zero is the slow eigenvalue, and it is more than 39 units above either faster eigenvalue.

2. Obtain its vector from determinants

For a square matrix C, its adjugate is the transpose of the array of signed minors. The identity C adj(C) = det(C)I follows by expanding determinants along a row. Apply this to $C = M^{T} - xI$. At an eigenvalue r, det(C) = 0, so any nonzero adjugate column is an eigenvector. The report prints four polynomial entries of such a column. Rational interval evaluation proves that its first entry is nonzero, allowing division to set $w_1 = 1$.

A polynomial can be evaluated using repeated multiplication and addition, for example $2x^2+3x+4 = (2x+3)x+4$. Replace x by an interval containing r and carry lower and upper bounds through those operations. This encloses each vector entry. It is not necessary to know r exactly as a decimal.

3. Enclose thermal weights using integer powers

To prove $L \le q^{1/n} \le U$ for positive numbers, it is enough to verify $L^n \le q \le U^n$. The certificate proposes nearby rational L and U, then accepts them only after exact power comparisons. Combining those intervals with the vector intervals encloses both overlaps.

$$-0.0937454280 < b_h < -0.0937454279,$$
$$-0.0938488357 < b_c < -0.0938488356.$$

Both intervals are negative and separated. Thus the colder preparation has strictly larger slow-mode magnitude. A separate checker reaches the same conclusion using matrix elimination and a bound on how far an approximate eigenvector can be from the true slow direction. The two methods provide different routes to the same claim.

The 11 manuscript comparisons check the matrix, cubic, cofactors, triplet table, root interval and printed overlap bounds. The standalone checker uses only Python's standard library.

▼ **When ordinary ordering is preserved at every time**

1. Follow adjacent differences

For values $f_1,\dots,f_n$ on energy-ordered states, define $y_i = f_{i+1}-f_i$. The function is increasing exactly when every $y_i \ge 0$. The generator $L = M^T$ acts on functions. Because L sends constant functions to zero, its action on differences is described by an (n–1)-by-(n–1) matrix K.

If every off-diagonal entry of K is nonnegative, the exponential-series argument from the contrast lesson gives $e^{tK} \ge 0$. Therefore an initially increasing function stays increasing after evolution. This property is called stochastic monotonicity. It is stronger than merely having the contrast matrix from Step 14 nonnegative off the diagonal: the two matrices use different coordinates.

For example, one required four-state entry is $K_{21} = a_{12}-a_{13}$. It equals –27 in both counterexamples here, so they do not have this ordering property. There is no conflict between them and a theorem that assumes it.

2. Translate function order into probability order

A probability distribution p is stochastically above q if the average of every increasing function is at least as large under p. For ordered discrete states, this says p has at least as much probability above every cutoff. Hotter Gibbs preparations have this order because increasing temperature shifts weight toward higher energies. A semigroup preserving increasing functions also preserves this order between distributions.

For a reversible process, the density ratio p_i/q_i evolves by the same function semigroup. On cooling, a Gibbs preparation divided by the bath distribution is increasing with energy, and it stays increasing. This is the extra fact needed to compare distances.

3. Use the supporting-line inequality for a convex function

A convex curve lies above each of its supporting lines. If g(y) is the slope of a supporting line at y, then $\Phi(x)-\Phi(y) \geq g(y)(x-y)$, and slopes can be chosen nondecreasing. Apply this to the density ratios of farther and nearer preparations and multiply by the bath probabilities:

$$D_\Phi(p_{far}\|q)-D_\Phi(p_{near}\|q)$$
$$\geq \Sigma_i (p_{far,i}-p_{near,i})\, g(p_{near,i}/q_i) \geq 0.$$

On cooling, the last slope vector is increasing, so stochastic order gives the last inequality. On heating, both relevant orders reverse, leaving the same conclusion. With $\Phi(x) = x \ln x$ this is KL divergence; with $\Phi(x) = |x-1|/2$ it is total variation. KL is strictly convex, and finite-time matrix evolution is invertible, so distinct preparations have strict KL ordering at every finite time.

This argument does not need a simple slow mode. The separate asymptotic proof does: a nonnegative eigenvector for K lifts to a monotone slow function, whose Gibbs covariance derivative has a strict sign.

The ordering principle has established precedents: Baake and Sumner, Proposition 3.11, and Ohga, Hayakawa and Ito, Result 2 and SI D2. The latter's stated no-Mpemba result concerns energy; the supporting-line derivation above supplies the convex-divergence conclusion under the assumptions stated here.

▼ **How the energies alone can rule out cooling in the larger contrast family**

1. A decreasing logarithmic slope cannot turn upward

Return to the contrast coordinates g_k from Step 14. On cooling $R_k = -g_k$ is positive. Apart from a fixed positive factor, it is a decaying exponential times

$$[\alpha_k - \Sigma_{i \le k}\, e^{\beta(E_{k+1}-E_i)}]/Z(\beta),$$
$$\alpha_k = \Sigma_{i \le k}\, e^{\beta_b(E_{k+1}-E_i)}.$$

The bracket is positive before the bath, with negative first and second derivatives. Its logarithm has negative second derivative. Also $(\ln Z)''$ is the positive energy variance. Thus $(\ln R_k)'' < 0$. If R_k starts with a nonpositive derivative at $\beta = 0$, its logarithmic slope becomes strictly negative for every later cooling β. This includes an initially zero derivative.

2. Reduce all starting slopes to one energy test

Shift E_1 to zero, put $a = E_2$, and let $\mu = (E_1+\cdots+E_n)/n$ be the ordinary arithmetic mean energy. For the first contrast, $g_1'(0)$ has the sign of $\mu - e^{\beta_b a}(\mu - a)$. If $\mu \le a$, this is positive. Otherwise it is nonnegative exactly when

$$\beta_b \le \beta* = \ln[\mu/(\mu-a)]/a.$$

This one condition controls every later contrast too. For $y = E_{k+1} \ge \mu$, the required starting slope is positive because a proper initial segment of the energy list has mean below μ. For $y < \mu$, compare averages of the increasing function $1/(\mu - t)$. Its average over $[E_i, y]$ is at least its average over $[0, a]$. Integrating gives

$$\ln[(\mu - E_i)/(\mu - y)]/(y - E_i) \ge \ln[\mu/(\mu-a)]/a.$$

Exponentiation now bounds each term in the starting slope in the required direction. All cooling contrast magnitudes therefore decrease toward the bath. The nonnegative contrast evolution preserves their componentwise order, excluding finite-time weighted squared-distance reversal. The slow-projection argument also excludes asymptotic KL reversal.

3. Why this threshold is exact for the stated family

If $\beta_b > \beta*$, the first contrast initially increases in magnitude. It can be made the simple slow mode by adding a small rank-one term to a reset generator. All G constraints then hold at equality, and finite positive preparations sufficiently near $\beta = 0$ witness cooling. This establishes sharpness for the G-only family. It does not assert that those rates satisfy the extra F constraints.

For four equally spaced energies, $\mu = 3a/2$, so the threshold is $\beta_b a \le \ln 3$. A spacing-independent consequence for $n \ge 3$ is $\beta_b(E_n - E_1) \le n/(n-2)$. In four states the latter cutoff is 2. The proof bounds μ by placing every energy above E_2 at the highest energy and expands the resulting logarithms; every correction beyond $n/(n-2)$ is positive. Distinct energies approaching that limiting arrangement show the constant cannot be improved for the G-only family.

4. The G-only amplitude limit is larger

Each contrast magnitude has a decreasing positive numerator. Its cold-to-hot ratio is therefore strictly less than the hot-to-cold denominator ratio. If the contrast compares the first $m-1$ states with state m, the first m terms of the cold denominator contribute at least m, and the other $n-m$ hot terms contribute at most one each. The ratio is below $n/m \le n/2$. The slow-mode ratio is a nonnegative weighted average of these contrast ratios, so it too is below $n/2$.

This bound can be approached. Use energies 0, 1/L, L, 2L,... and the first contrast as the slow mode. With $\beta_h = L^{-2}$ and $\beta_c = L^{-1/2}$, the hot partition function tends to n and the cold one tends to 2; the two overlap numerators are both asymptotic to $-1/L$. The ratio tends to $n/2$. Thus the G-only amplitude supremum is $1-2/n$, or 50% for four states. The full F-and-G premise gives the stronger 25% result proved earlier.

▼ **Why the general four-state test needs only endpoints and one possible extra point**

1. Turn four derivatives into a curve in a plane

Use energies $(0,a,b,c)$, with $0 < a < b < c$. The ground-state probability increases with β, so $g = \pi_1' > 0$. Define $u = \pi_2'/g$ and $v = \pi_3'/g$. Since the four probability derivatives sum to zero, any fixed mode w has

$$b'(\beta) = g(\beta)\Phi(\beta),$$
$$\Phi = A+Bu+Cv,$$
$$A = w_1 - w_4, \quad B = w_2 - w_4, \quad C = w_3 - w_4.$$

Only the sign of Φ matters. As β varies, (u,v) traces a curve. Exact differentiation shows u decreases strictly and v, considered as a function of u, has strictly positive curvature. Therefore a line $A+Bu+Cv = 0$ intersects it at most twice, counting a tangency twice.

2. Check the determinant signs behind that claim

A Wronskian is a determinant whose rows contain functions and their successive derivatives. It measures whether those functions can locally mimic one another. For g, gu and gv, elementary determinant operations give

$$W(g,gu) = g^2u',$$
$$W(g,gu,gv) = g^3(u')^3\, d^2v/du^2.$$

Both determinants are negative for four ordered energies. To see the signs without trusting a graph, first add the constant function to replace derivatives of probabilities by probabilities. Multiply each column function by the positive partition function Z; the product rule changes the determinant by the positive factor Z to the number of columns. The columns now contain simple exponentials. Their derivatives multiply them by powers of $-E_i$, producing a Vandermonde determinant: a product of energy differences. Expanding the Z column leaves only positive exponential weights times those signed products. The two required products have negative signs. Hence $u' < 0$ and $d^2v/du^2 > 0$.

3. Find the only possible interior extremum of Φ

Put $c_0 = ab(b-a)$, $c_1 = ac(c-a)$, $c_2 = bc(c-b)$, $\gamma = (c-a)/(c-b) > 1$, and $x = e^{-(c-b)\beta}$. The slope of the curve is

$$s(x) = dv/du = (c_2x^\gamma - c_0)/(c_0+c_1x).$$

Its derivative has positive numerator $c_0c_2\gamma x^{\gamma-1}+c_1c_2(\gamma-1)x^\gamma+c_0c_1$. Thus s increases strictly with x, from $s(0) = -1$ to $s(1)$. If $C \neq 0$, an interior extremum of Φ must solve $s(x^*) = -B/C$. There is exactly one such solution when $-1 < -B/C < s(1)$, and none otherwise. Equality at either endpoint is excluded. When $C = 0$, Φ is already monotone or constant.

For example, energies (0,1,2,3) give $s(x) = (6x^2-2)/(2+6x)$. If $-B/C = -1/3$, the unique solution is $x^* = 1/3$ and $\beta^* = \ln 3$. One scalar equation replaces an unlimited temperature scan.

4. Test signs, not just zeros

On cooling, evaluate Φ at $\beta = 0$, at the bath, and at β^* if it lies strictly between them. On heating, evaluate it at the bath, at infinity using $\Phi(\infty) = w_1 - w_2$, and at β^* if it lies on that side. A side has an ordinary asymptotic reversal exactly when its list contains both a strictly positive and a strictly negative value.

Why both signs? If b′ keeps one sign, b is monotone and its zero at the bath forces normal magnitude order. If b′ takes both signs on cooling, consider its last sign-changing zero before the bath. Just after that zero, b must head toward its bath value zero; just before it, b has the same nonzero sign but its derivative has the opposite sign. Its magnitude is therefore increasing there. The heating proof uses the first sign-changing zero after the bath. This produces a finite interval containing actual finite-temperature witnesses.

A tangent zero without opposite signs is insufficient. A zero exactly at $\beta = 0$ or infinity is a limiting statement, not a finite preparation. For a strong effect, additionally find zeros of b itself: split at the stationary points, check each strictly monotone interval for opposite endpoint signs, and count stationary zero points separately. General transcendental equalities require proof or rigorous enclosures; floating-point tolerances do not decide them.

See the full classification proof, three exact determinant checks, and detailed review. The rate construction used to realize prescribed modes has precedent in Klich and colleagues, Appendix VIII.

▼ **When several slow patterns tie: project onto the whole space**

1. Replace one coefficient by a vector

Suppose two independent patterns both decay as $e^{\lambda t}$. Rotating the coordinate axes within their plane changes the individual coefficients, but not the total slow component. A vector (3,4) can become (5,0) after a rotation; the zero coordinate does not mean the vector disappeared. The same issue occurs with a repeated slow eigenvalue.

Divide each probability deviation by $\sqrt{q_i}$, where q is the bath distribution. Detailed balance makes the generator symmetric in these coordinates. Let P be the orthogonal projection onto its entire slow eigenspace. The relevant nonnegative amplitude is

$$Q(\beta) = \|P\, D^{-1/2}(\pi(\beta)-q)\|^2 = \pi(\beta)^{\mathsf{T}} H \pi(\beta),$$
$$H = D^{-1/2} P\, D^{-1/2}.$$

The bath disappears because $P\sqrt{q} = 0$. The leading chi-square distance is $e^{2\lambda t} Q(\beta)$, and leading KL is half of it. A strong preparation requires $Q = 0$, meaning the entire projected vector vanishes. Equal nonzero Q values for a specified pair leave the subleading terms to decide their ordering.

2. Count how many thermal preparations can disappear

With n distinct energies, any k ≤ n distinct Gibbs vectors are linearly independent. To see why, multiply each by its positive partition function. A hypothetical dependence would make a nonzero combination of k distinct exponentials vanish at k different temperatures. Divide by one exponential and differentiate: one term disappears. Repeating Rolle's theorem gives at most k–1 zeros, a contradiction.

If the slow space has dimension r, the kernel of its projection has dimension n–r. It already contains the transformed bath vector. It can therefore contain at most n–r–1 additional independent Gibbs vectors. For four states, the maximum numbers of nonbath strong preparations are 2, 1, and 0 for slow multiplicities 1, 2, and 3.

3. Why the bound can be reached

Choose the desired preparations and span them together with the bath. Project onto the perpendicular complement of this span, then use $S = -I + ss^{\mathsf{T}} + \varepsilon P$, with $s = \sqrt{q}$ and a sufficiently small positive ε. Its slow space is exactly the chosen complement. Small ε preserves positive transition rates because the unperturbed reset has strictly positive rates. This construction establishes the bound in the unrestricted reversible family; it does not promise the triplet-failure premise.

4. The full triplet premise is much more restrictive

If n ≥ 4 and only one nonstationary mode is faster than all the others, the full F-and-G conditions force the generator to be a reset plus mixing on one adjacent pair. The fast vector is supported only on that pair. The proof uses the last nonzero component to order the earlier components with G, then F forces all but its adjacent predecessor to vanish.

For four states this covers all repeated-slow cases, and stochastic order gives normal convex-divergence ordering at every time. For five or more states, smaller repeated slow spaces can have multiple faster modes; the same classification cannot be assumed. An exact five-state example has a negative adjacent-difference matrix entry despite all F and G being nonpositive.

Full arguments: slow-space dimension and one-fast-mode classification. The step-by-step review records the assumptions and dimension exceptions.

▼ **A finite-time crossing can disappear again: solve the three-exponential test**

1. Subtract the two squared distances

In the symmetric coordinates, different eigenspaces are perpendicular. Squaring the length therefore gives no cross terms. Group repeated eigenvalues together and subtract the nearer preparation from the

farther one:

$$\Delta(t) = \sum_j A_j \, e^{2\mu_j t},$$
$$A_j = \|P_j z_{far}\|^2 - \|P_j z_{near}\|^2.$$

There are at most three distinct nonzero eigenvalues in four states. A coefficient A_j can vanish even when both individual projections are nonzero; discard that exact zero before identifying the dominant remaining exponential. For two different Gibbs preparations on the same bath side, $\Delta(0)$ is strictly positive.

2. Factor out a positive exponential

With three remaining terms the sign is the sign of $F(t) = A + Be^{-at} + Ce^{-bt}$, where $0 < a < b$. Differentiating gives $F' = -aBe^{-at} - bCe^{-bt}$. A positive stationary time exists only when $BC < 0$ and $-bC/(aB) > 1$. If it exists, it is

$$t* = \ln[-bC/(aB)] \,/\, (b-a).$$

Check $F(0)$, the limiting constant A, and $F(t^*)$ when that time is admissible. A negative minimum proves a reversal. A zero minimum does not: the graph may only touch zero or approach it forever. With two terms, the factored expression is monotone, so only endpoints are needed.

3. A hand-solvable illustration

Let $x = e^{-t}$ and $F = 1-6x+6x^2$. At $t = 0$, $x = 1$ and $F = 1$. At very late times x tends to zero and F tends to 1 again. But at $x = 1/2$, $F = -1/2$. Solving the quadratic gives $x = (3 \pm \sqrt{3})/6$. Both roots lie strictly between zero and one, producing two finite crossings and a reversed interval between them.

This little polynomial illustrates the sign pattern; it is not by itself a physical generator. The report's separate exact construction supplies positive reversible rates, genuine thermal preparations, and rational spectral coefficients with precisely this pattern. Its crossing distances are extremely small, so it demonstrates a mathematical possibility rather than an engineering benefit.

See the finite-time theorem and its exact transient construction. The test concerns weighted Euclidean distance and its square; KL and total variation require their own finite-time analysis.

▼ **Can a proposed slow pattern come from allowed rates?**

1. An attractive curve is not yet a realizable model

You can write down values $w_1,\ldots,w_4$ that produce an interesting thermal overlap. To realize the pattern, it must be an eigenfunction of a probability-conserving reversible matrix, be its simple slowest nonstationary mode, and satisfy all the rate constraints. The eigenvalue equation alone does not guarantee the last two requirements.

2. Design the two faster directions first

Subtract the bath mean from w, so $q^{\mathsf{T}}w = 0$. Choose two independent columns U spanning the directions perpendicular to both the constant function and w in the bath-weighted inner product. For a symmetric two-by-two matrix B, form the signed auxiliary matrix $M_0 = -DUBU^{\mathsf{T}}$. It kills the stationary and proposed slow directions. If B is positive definite, its other two directions have strictly negative eigenvalues.

This auxiliary matrix may have negative off-diagonal entries. Add a sufficiently large reset $\rho(q1^{\mathsf{T}}-I)$. That makes all transition rates positive and shifts all nonstationary eigenvalues by $-\rho$. The proposed w now has eigenvalue $-\rho$, strictly slower than the other two. Reset addition leaves every F and G unchanged.

3. The remaining problem is a disk and straight lines

Positive rescaling lets us set trace B = 1:

$$B = \tfrac{1}{2}\begin{bmatrix} 1+x & y \\ y & 1-x \end{bmatrix}.$$

Its eigenvalues are $(1 \pm \sqrt{x^2+y^2})/2$.

Both are positive exactly when $x^2+y^2 < 1$.

The eight F-and-G inequalities are linear in x and y: each cuts out a closed half-plane. Find their common point closest to the origin. It must be the origin, a perpendicular projection onto one boundary line, or an intersection of two nonparallel boundary lines. These are finitely many candidates. With rational q and w, their coordinates and squared lengths are rational.

A feasible candidate with squared length strictly below one realizes the mode. Equality to one does not: one fast gap vanishes and the proposed slow mode is no longer simple. A test that rounds 1 to 0.999999 would change the mathematical problem. The regression suite includes exact boundary rejection.

4. Necessary conditions need not be sufficient

The inequalities $w_1 \geq w_2$, $w_1+w_2 \geq 2w_3$, and $w_4 \leq 0$ are necessary under the full premise. They do not replace the disk test. The report gives a vector satisfying them strictly but having a positive weighted sum of three required inequalities equal to trace B. The inequalities would force trace $B \leq 0$, contradicting positive definiteness. This is an explicit impossibility certificate.

See the feasibility proof, exact constructor, and separating identity.

▼ **Why one endpoint decides cooling under the full triplet premise**

1. Keep the premises visible

This shortcut assumes all four original triplet predicates fail and the slow mode is simple. It does not apply to an arbitrary four-state mode. In the contrast orientation the cooling overlap b is negative, and b' at the bath is strictly positive.

2. Split according to w_3-w_4

If $C = w_3-w_4$ is nonnegative, the conditional first-three mean is increasing and stays above w_4. Differentiating $b = Pm+(1-P)w_4$ gives $b' = Pm'+P'(m-w_4) > 0$, so there is no cooling anomaly.

If $C < 0$, use the derivative curve from the preceding explanation: $b' = g\Phi$ with $g > 0$ and $\Phi = A+Bu+Cv(u)$. Because $v(u)$ is strictly convex, Φ is strictly concave. At the low-temperature endpoint its limit is $w_1-w_2 \geq 0$. At the bath it is positive. If its infinite-temperature-end value is also nonnegative, the whole interior lies above the nonnegative chord and has positive derivative.

If that last endpoint value is negative, concavity permits exactly one crossing before reaching the bath. Thus b' is negative first, then positive forever. Since b is negative on cooling, its absolute value first grows and then shrinks. This gives a unique maximum and finite hot preparations with reversed ordering.

$$\text{Cooling exists} \Leftrightarrow b'(0) < 0 \Leftrightarrow \mathrm{Cov}_{\mathrm{uniform}}(w,E) > 0.$$

Here $\beta = 0$ is an analytic endpoint used to decide the sign. It is not presented as a finite-temperature witness. Strict negativity persists at nearby positive β. Endpoint equality instead gives strict normal ordering at every finite positive β.

3. Avoid choosing an eigenvector by using a projector

The report computes the same sign using the largest nonzero cubic root r and a scalar quadratic $\mathscr{R}(r)$ built from M, M^2, and the bath probabilities. The key identity is

$$\mathscr{R}(r) / p'(r) = b(0)b'(0) / (w^{\mathsf{T}}Dw).$$

The denominator on the right is positive, $p'(r) > 0$, and $b(0) < 0$. Therefore $\mathscr{R}(r) > 0$ means cooling. The polynomial projector is used only on zero-total displacements; it need not annihilate the stationary direction on the full space. If r is repeated, the separate repeated-mode theorem applies instead of dividing by $p'(r) = 0$.

The endpoint proof and rate evaluation show each identity and sign.

▼ **How exact algorithms decide signs and zeros without a numerical tolerance**

1. Turn commensurate energies into polynomials

If $E_i = m_i\Delta$ with integer m_i, set $x = e^{-\beta\Delta}$. Thermal weights become x^{m_i}. After shifting all exponents equally, the partition function Z and slow numerator N are polynomials. The derivative sign is the sign of $N'Z–NZ'$ for a simple mode, or $N'Z–2NZ'$ for the squared projection amplitude.

Factor this polynomial by multiplicity. A root of odd multiplicity changes sign; a root of even multiplicity merely touches zero. Count the odd roots on the open cooling and heating intervals. Count all distinct numerator roots separately for strong preparations. Remove the bath and the zero/infinite-temperature endpoint factors exactly.

2. Algebraic coefficients need the correct real root

A cubic can define several real numbers. Saying “α is a root” is insufficient: an isolating rational interval selects which one. A polynomial expression in α may have a negative value even if its leading coefficient is positive. For example, the real root α of $\alpha^3–\alpha–1 = 0$ lies between 1 and 2, so $\alpha–2$ is negative.

The implementation first reduces expressions exactly to detect zero. For a nonzero expression it evaluates rational intervals and refines the selected root interval until zero is excluded. The Sturm sequence counts roots using these correctly embedded signs. It does not infer signs from the appearance of a coefficient list.

3. Incommensurate rational weights lead to prime logarithms

With arbitrary rational bath weights, the full-premise endpoint scalar is a sum of algebraic coefficients times logarithms of rational numbers. Factor those rationals into primes to write $\mathscr{R}(r) = \Sigma \gamma_p \ln p$. Unique prime factorization makes the prime logarithms independent over rational coefficients. The cited form of Baker's theorem then makes them linearly independent over algebraic coefficients. Consequently the sum is zero exactly when every γ_p is zero. This uses an established theorem; it does not assert algebraic independence of logarithms.

If some coefficient is nonzero, convergent rational logarithm bounds eventually determine the sign. Range reduction keeps the series arguments small. A resource cap can stop refinement early, but the returned answer must then be "unresolved," not "no cooling." The tests exercise both exact equality and small opposite signs that need more than one refinement round.

Read the polynomial reduction and prime-logarithm proof. Their rationality and triplet assumptions are substantive, not implementation conveniences that may be dropped.

▼ **A real audit finding: a root near the bath was easy for a reference test to miss**

1. The problematic shortcut

A supporting test compared the numerical classifier against signs sampled between rational root-enclosing intervals. It retained a root only when the whole enclosure lay strictly inside the temperature interval. An enclosure touching the bath boundary does not prove that its root lies on the boundary: the root can be slightly inside.

2. Reproduce the failure with a thermal polynomial

For $Z = 1+x+x^2+x^3$, choose $N = x+(-8/17+\varepsilon)x^3$ with $\varepsilon = 10^{-30}$. Its thermal derivative numerator $H = N'Z - NZ'$ satisfies

$$H(1/2) = 17\varepsilon/16 > 0,$$
$$H(1) = -82/17+6\varepsilon < 0.$$

So an interior sign change exists between $x = 1/2$ and $x = 1$. But a finite-width root enclosure begins exactly at 1/2 and extends only slightly above it. The old reference skipped that root and sampled a negative value farther inside, missing the tiny positive interval.

3. Repair the reference, then challenge it

The reference now counts odd-multiplicity roots with exact rational arithmetic after exactly removing endpoint factors. It finds the missing root. Thirteen regression checks cover this example, true endpoint roots, even tangencies, repeated odd roots, roots near zero and one, and algebraic coefficient fields. The main exact classifier already used exact endpoint removal and passes this new example unchanged.

This was a defect in a supporting audit reference, not a refutation of the mathematical theorem or a demonstrated error in the production exact classifier. Recording it matters: a checker needs scrutiny too.

Inspect the finding and regression results, repaired reference, and reproduction test.

▼ **Why an ordered chain of contrasts preserves every convex heating distance**

1. Start with an average that cannot be negative

If two quantities increase together, their covariance is nonnegative. For values a_i and an increasing function g, with positive weights p_i summing to one, expand the two sides to check

$$\operatorname{Cov}(a,g(a)) = \tfrac{1}{2}\sum_i\sum_j p_i p_j (a_i-a_j)(g(a_i)-g(a_j)) \geq 0.$$

Each product is nonnegative because its two factors have the same sign. For a convex differentiable function Φ, the slope Φ' is increasing, so we can use $g = \Phi'$. Strict convexity makes the covariance positive whenever at least two a values differ.

2. Apply that fact to a prefix of physical states

Write the density ratio $r_i = p_i/q_i$ as $r = 1+Hx$, where H contains the unnormalized prefix contrasts. On heating, all x coordinates are positive. Put $g_k = \sum_i q_i H_{ik} \Phi'(r_i)$; this is the derivative of the divergence with respect to coordinate x_k. Individual g_k need not be positive. Instead look at cumulative sums $S_k = \sum_{j \leq k} x_j g_j$.

Higher contrasts are constant on states 1 through k+1. Their combined contribution there is $c_k = 1+\sum_{j>k} x_j$. The lower contrasts have zero weighted mean on that prefix. Therefore

$$S_k = \sum_{i \leq k+1} q_i (r_i - c_k)\Phi'(r_i) \geq 0.$$

This is exactly prefix mass times the covariance just derived. Notice the distinction: cumulative sums are positive even when individual terms can be negative.

3. Learn the small summation trick that uses those prefixes

Let $y = \partial x/\partial\beta$ and $R_k = y_k/x_k$. If $R_1 \geq R_2 \geq \ldots > 0$, then collecting the coefficient of each $x_k g_k$ gives

$$\Sigma_k R_k x_k g_k = R_{n-1} S_{n-1} + \Sigma_{k<n-1} (R_k - R_{k+1}) S_k \geq 0.$$

This is summation by parts. For three terms, expand $R_3S_3+(R_2-R_3)S_2+(R_1-R_2)S_1$ to recover the original sum. The identity turns information about partial sums into information about the weighted total.

4. Explain why the ratios keep their order

Initially each x_k is proportional to δ times an integral of $Q(s)e^{-\delta s}$, truncated at E_{k+1} and divided by the relative partition function. Here $\delta = \beta - \beta_b > 0$ and $Q(s)$ is the bath mass below energy s. Differentiating its logarithm gives a common term minus the mean s in that truncated integral. Extending the upper endpoint increases that mean, so R_k decreases with k.

A tridiagonal Metzler contrast generator only couples neighboring contrast coordinates, with nonnegative off-diagonal entries. Its exponential has nonnegative ordered 2×2 determinants. The determinant expansion then preserves decreasing y_k/x_k ratios. One can prove the determinant claim by differentiating those determinants: their evolution again has nonnegative off-diagonal coefficients because a neighboring index cannot cross another occupied index.

This condition concerns *contrast coordinates*, not neighboring physical energy states. It is also a real restriction: a general G-admissible generator can destroy the ratio order. The resulting proof-route failure is not itself a KL reversal.

5. Keep track of strictness and nonsmooth distances

At finite time a distinct initial state cannot become exactly equilibrium, because a matrix exponential is invertible. Thus the full covariance is positive for strictly convex Φ, and the last R is positive. Integrating the positive thermal derivative proves strict heating order. For nonsmooth convex functions, ordered subgradients replace Φ' almost everywhere; local Lipschitz continuity permits integration. Non-strict convexity gives the non-strict conclusion.

See the full argument and the classical determinant result in Schwarz (1970), Theorem 2. The argument above explains the project's application; the determinant theorem is established mathematics.

▼ **A guaranteed KL window: turn temperature distance into an angle estimate**

1. Form a dimensionless measure of how far you start from the bath

Let $\delta = |\beta - \beta_b|$ and $c = E_{max} - E_{min}$. The product $z = \delta c$ is dimensionless: inverse temperature times energy. Small z means the Gibbs weights have not spread very far relative to the bath. The result guarantees ordinary KL ordering on either side whenever $z \le 2/\sqrt{5}$, approximately 0.8944. Cooling still requires positive initial β.

2. Compare a vector with the direction in which it is growing

After choosing the outward sign, write the positive contrast coordinates as x and their outward thermal derivative as y. Integral formulas for the initial Gibbs weights give, coordinate by coordinate,

$$(1/\delta - c)x \le y \le (1/\delta + c)x.$$

The nonnegative contrast semigroup preserves both inequalities at every time. Thus y points broadly in the same direction as x. More precisely, if $a = 1/\delta - c$ and $b = 1/\delta + c$, the cosine of their angle is at least $2\sqrt{(ab)}/(a+b) = \sqrt{(1-z^2)}$, provided $z < 1$.

To derive this without guessing an angle, put $v_i = y_i/x_i$ and assign weight $x_i^2/\|x\|^2$. Since $a \le v_i \le b$, the inequality $(v_i - a)(v_i - b) \le 0$ gives $E[v^2] \le (a+b)E[v] - ab$. The squared cosine is $E[v]^2/E[v^2]$. Its lower bound reduces to the nonnegative square $((a+b)E[v] - 2ab)^2$.

3. A positive matrix can change the angle, but only by a bounded amount

KL is curved, rather than exactly a squared Euclidean distance. Its Hessian describes that curvature. Along the straight line from equilibrium to the evolved density r, the integrated Hessian T has all eigenvalues between $m = 1/r_{max}$ and $M = 1/r_{min}$. The initial density range has $r_{max}/r_{min} = e^z$, and Markov averaging keeps the evolved density inside that range.

The outward KL derivative is $y^T T x$. Split T into its scalar center $(M+m)I/2$ and a remainder of norm at most $(M-m)/2$. The scalar part rewards alignment; the remainder can reduce the result only by its norm times $\|x\|\|y\|$. Hence

$$\partial D_{KL}/\partial\delta \geq \tfrac{1}{2}(M+m)[\sqrt{(1-z^2)}-\tanh(z/2)]\|x\|\|y\|.$$

4. Check the advertised endpoint, rather than assuming a small-parameter approximation

For $0 < z \leq 2/\sqrt{5}$, squaring the nonnegative sides gives $\sqrt{(1-z^2)} \geq z/2$. Also $\tanh(u) < u$ for positive u, so the bracket is strictly positive even at the upper endpoint. Both vectors remain nonzero at every finite time. The bath endpoint itself has zero KL; it does not require substituting $\delta = 0$ into $1/\delta$.

This is an exact sufficient bound, uniform in time, rather than a truncated Taylor expansion. Repeated eigenvalues do not invalidate it. Beyond this window the proof becomes inconclusive; it does not predict a reversal. The numerical audit supports the proof on sampled cases, but its finite samples cannot establish the universal theorem.

Read the full theorem, exact algebra checks, and sampling results with exclusions.

▼ **What is proved about higher dimensions, and what remains a conjecture?**

1. Combine contributions only after checking their signs

The contrast theorem writes the oriented slow mode as a nonnegative combination of prefix contrasts. Group its first two contrasts into f, supported on the first three states. If the additional inequality $w_1+w_2 \geq 2w_3$ holds, f has a conditional thermal average that strictly increases toward its bath value zero. On cooling that average is negative.

Its full overlap is the conditional mean times the probability P_3 of being in the first three states. At finite positive β, $3/n < P_3 < 1$. Therefore its cold-to-hot magnitude ratio is less than n/3: the conditional magnitude shrinks and the probability factor grows by less than n/3.

Each later contrast similarly has a ratio less than n/(k+1), hence at most the stronger threshold n/4. All nonzero components are negative on cooling, so no cancellation occurs. The ratio of their sum is a weighted average of their ratios, with hot magnitudes as positive weights. It is consequently below n/3.

```
Additional arithmetic inequality ⇒ amplitude advantage < 1−3/n.
For n > 4, the full triplet premise has not yet been proved to imply
that additional inequality.
```

2. One special tail can be compressed exactly for this purpose

Suppose the slow mode takes one constant value on states 4 through n. Let Q_T be their total bath probability, and require $Q_T < q_3$. Average the rates out of this tail using its conditional bath probabilities, and sum the reverse rates. This gives a four-state reversible compression with ordered weights (q_1, q_2, q_3, Q_T).

The constant-tail eigenfunction survives the compression. The variational bound on eigenvalues keeps it simple and slow. Direct rate identities preserve every F and G inequality: the new F is an average of old F values plus nonnegative multiples of old nonpositive G values. The established four-state theorem then supplies the missing arithmetic inequality.

Full dynamic lumpability is unnecessary because only this eigenfunction is being retained. But a constant tail and a sufficiently small *total* tail weight are essential assumptions. This averaging is different from eliminating a state through a Schur complement.

3. Convert a proposed mode into a graph construction

Fix positive contrast coordinates x. Connect contrast coordinates i and j with symmetric nonnegative weights ℓ_{ij}. Choose the diagonal of B to make $Bx = 0$. The central identity is

$$-y^{\top}\eta B y = \Sigma_{i<j}\, \ell_{ij} x_i x_j (y_i/x_i - y_j/x_j)^2 \geq 0.$$

Here η records the squared lengths of the unnormalized contrast columns. The right side resembles a network's energy: it vanishes exactly when y_i/x_i is constant on each connected component. For a connected graph, the kernel therefore has only the one direction x. All other directions decay.

Transform B back to physical coordinates. Some rates may initially be negative. Adding a sufficiently large reset makes them positive, shifts every nonstationary eigenvalue equally, and leaves every F and G test unchanged. For fixed bath weights and x, the triplet tests are linear in ℓ, so a linear program can ask whether all graph edges and test margins can be strictly positive together.

4. Understand the exact scope of this decision reduction

A positive optimum realizes the prescribed mode under strict tests. A failed strict program does not reject every fixed mode on a weak boundary: there connectivity, rather than positivity of every edge, is the right condition. A global strict violation of the arithmetic inequality could be perturbed into the strict case and represented by rational construction data.

This turns the open question into a precise feasibility problem. It does not solve that problem for all inputs. Nor would an arithmetic counterexample alone establish a larger thermal advantage; actual Gibbs overlaps would still have to violate the proposed bound.

Sources: conditional and rare-tail reductions, graph feasibility theorem, and the 66-step review.

▼ **Add one state without losing the counterexample**

1. Give the new state a very small bath weight

Suppose an n-state model already has a simple slow mode and a strict cooling reversal. Choose a new highest energy E_{new}, with bath weight $\varepsilon = e^{-E_{new}}$. Small positive ε means a large but finite energy. From the new state, allow transitions to every old state at the same positive rate A. Detailed balance determines the reverse rates:

```
new → i: A,
i → new: Aε/qᵢ.
```

Here q_i are the old unnormalized bath weights and the bath inverse temperature is one. Keep all original off-diagonal rates. Recompute the diagonal losses because each old state now has an extra way to leave.

2. Check the new triplets directly

An old triplet is unchanged. A new triplet consists of old states $i < j$ and the highest state. Put $t = q_j/q_i < 1$, $u = \varepsilon/q_j > 0$, and C equal to the old downward $j \to i$ rate. Substitution into the original tests gives

```
G = A(t−1) < 0,
F = −(1−t)(C+Au/2) < 0.
```

All factors have explicit signs. This is why an equal-rate connection to the new state preserves the required triplet failures.

3. Keep the new relaxation mode fast

As ε tends to zero, upward transitions disappear. The new state empties at total rate nA. Choose A large enough that its decay rate −nA is below every original nonzero eigenvalue. The limiting spectrum then contains all old rates plus one faster rate. The old slow eigenvalue stays isolated.

If w is the old left slow mode with eigenvalue λ, its added coordinate in this limit is $A\Sigma_i w_i/(\lambda+nA)$. This follows by writing the new-state eigenvector equation and solving for one unknown. Because the slow eigenvalue is simple, small changes in ε produce small changes in its eigenvalue and a consistently normalized eigenvector.

4. Keep the same two finite preparations

At a fixed positive inverse temperature β, the new thermal weight is ε^{β}, which tends to zero. Each preparation therefore approaches its old probability vector, with almost no probability on the added state. The overlaps approach their old values, so a strict reversal survives for sufficiently small positive ε.

The order of choices matters: fix the two positive β values first, then make ε small. At β = 0 the new weight would equal one, so this limiting argument would fail. Repeating the construction proves persistence in every larger dimension; it does not promise a larger advantage.

5. Handle an extremely rare coordinate carefully in the exact example

The finite five-state certificate uses $\varepsilon = 10^{-9000}$ and A = 1000. Its fifth thermal weights are exactly 10^{-9} and 10^{-90} at β = 1/1000 and 1/100. Exact matrix elimination brackets all four negative eigenvalues.

A small weighted eigenvector error alone would poorly control the rare fifth coordinate, because dividing by $\sqrt{\varepsilon}$ is enormous. The checker instead controls the four old coordinates and uses the fifth row equation to control the fifth separately. It proves a uniform overlap error below 10^{-6} and a strict hot-minus-cold overlap difference above 10^{-4}, with both overlaps negative. That difference uses the chosen mode normalization; it is not a percentage time saving.

Read the extension proof and exact five-state checker.

▼ **Why more states allow amplitude advantages approaching 100%**

1. The arithmetic idea: divide the same numerator by different state counts

In the four-state limiting family, the first three slow-mode components add to −2/15. A very hot preparation spreads its probability across all four states, giving overlap near (−2/15)/4. A less-hot preparation barely populates the highest state and effectively divides by three, giving (−2/15)/3. Their magnitude ratio tends to 3/4.

Replace the highest state by m = n−3 distinct rare states. If the hotter preparation spreads across all n states while the other mainly occupies the first three, the same calculation suggests

$$b_{hot} \to -2/(15n),$$
$$b_{cold} \to -2/45,$$
$$1-|b_{hot}|/|b_{cold}| \to 1-3/n.$$

For n = 4, 5, and 6, the achievable limiting amplitude advantages are 25%, 40%, and 50%. For n = 30 the limit is 90%. The calculation suggests a construction; the next steps establish that its rates and triplet tests are valid.

2. Split the rare state while preserving its aggregate motion

Choose positive distinct fractions r_j summing to one and give clone j bath weight $q_4 r_j$. Give every clone the old fourth state's rates to the first three states. Split reverse rates in proportions r_j. Inside the clone block, use reset rates $4r_j$ into clone j.

Summing the clone probabilities exactly recovers the old four-state evolution. Differences inside the clone block decay with rate $-4-\Sigma_i a_{i4}$, which is faster than −4. The old simple slow rate −3 remains, with a left slow mode that has the same value on every clone.

Thermal preparations away from the bath need not distribute clone probabilities in the proportions r_j. They may excite the extra modes. Those modes are faster, so this does not invalidate the asymptotic slow-overlap calculation.

3. Repair the inequality that splitting can break

A triplet with two low states and one clone has unchanged G, but its F changes by $(r_j - 1)q_4 G/(2q_{low})$. Both $r_j - 1$ and G are nonpositive, so this correction can be positive. Exact aggregation alone does not preserve all the tests.

The proof bounds the possible defect by $(5/3)\varepsilon^2$, using exact bounds on the original rates. It then adds $6\varepsilon^2$ times an auxiliary reversible generator H. For the troublesome triplets, H has $F \le -1/3$. The repaired bound is therefore

$$F_{repaired} \le (5/3)\varepsilon^2 - 6\varepsilon^2/3 = -\varepsilon^2/3 < 0.$$

The other triplet patterns either retain negative tests or gain strict negativity. All rates remain positive. The auxiliary generator uses constant downward rates inside the clone block; replacing them with constant reduced conductances would change this calculation.

4. Prove that the repair is small in the right sense

Some reduced conductances grow as the clone weights shrink. The proof therefore bounds outgoing physical rates, then uses reversibility to bound the symmetric matrix's operator norm. For each fixed n, the perturbation is of order ε^2. A spectral gap initially larger than 1/3 consequently stays larger than 1/4 for sufficiently small ε.

The weighted slow-vector error is of order ε^2. The smallest bath weight is of order ε^2, so conversion to a bound on individual components costs one factor $1/\varepsilon$, leaving an error of order ε. Any probability-weighted average changes by at most the largest component error. Hence all thermal overlaps retain their limiting values.

5. Take the limits in the stated order

Fix n and the clone fractions. Set $L = \ln(1/\varepsilon)$, $\beta_{hot} = L^{-2}$, and $\beta_{cold} = L^{-1/2}$. As ε shrinks, every hot thermal weight tends to one. Each cold clone weight contains $e^{-2\sqrt{L}}$ and tends to zero, while the three low weights tend to one. This justifies the first calculation, with finite positive temperatures at every chosen parameter.

To approach 100% amplitude advantage, first choose a sufficiently large finite n, then a sufficiently small positive ε for that n. The proof does not attain 100% in one finite model and does not establish a sharp upper bound for any fixed $n > 4$. The corresponding leading KL advantage tends to $1 - 9/n^2$ because its leading coefficient involves the squared amplitude.

6. Separate the finite example from the limiting theorem

A five-state generator at $\varepsilon = 1/1000$ is checked exactly for rates, triplets, and spectral separation, but is not itself the reported thermal witness. The separate witness uses $\varepsilon = 10^{-50}$, $\beta_{hot} = 1/10000$, and $\beta_{cold} = 1/50$. Rational thermal enclosures plus an explicit eigenvector-error bound place its repaired amplitude advantage between 37.6778168708% and 37.6778168731%.

These percentages describe slow-amplitude suppression. A useful finite-time cooling protocol would require its own distance, threshold, time, preparation-cost, and physical-realization analysis.

Sources: full cloning proof, rate and perturbation bounds, finite thermal certificate, and the 47-step review.

▼ **How a computer proves a whole parameter box instead of sampling it**

1. Replace a guessed number with an enclosure

A floating-point calculation returns a nearby representable number. An interval calculation returns two numbers guaranteed to lie below and above the exact answer. For example, if $x \in [2,3]$ and $y \in [4,5]$, then $x+y \in [6,8]$ and $xy \in [8,15]$. The guarantee concerns every allowed pair, including pairs you never sampled.

Computer endpoints themselves are rounded. The implementation widens each basic arithmetic result to the neighboring representable value in the outward direction. It uses MPFR directed rounding for decimal inputs, exponentials, logarithms, and square roots. These guarantees rely on the stated arithmetic environment and compilation settings; the compiler is instructed not to use fast-math or fused contraction.

A denominator interval containing zero does not justify division. A logarithm interval reaching zero does not justify taking a logarithm. Those cases become unresolved calculations that need narrower cells. They never become successful certificates by ignoring the invalid part.

2. Understand why a valid enclosure can still be too wide

If $x \in [0,1]$, ordinary interval subtraction gives $x-x \in [-1,1]$, although the true result is exactly zero. The enclosure is valid but loses the fact that both appearances represent the same x. This dependency problem can make a correct method fail to certify a true inequality.

The physical checker reduces this overestimation in several ways. It transforms constant edge matrices before multiplying uncertain rates, preserving useful cancellations. It also computes a value at the box center and adds derivative bounds times coordinate displacements:

$$f(\text{box}) \subseteq f(\text{center}) + \Sigma_k\, [\partial f/\partial x_k](\text{box}) \cdot (x_k - \text{center}_k).$$

This is the multivariable mean-value bound. Intersecting it with a direct interval evaluation is safe when both enclose the same function. A narrower intersection is stronger evidence; an empty intersection is an error or unresolved calculation, never proof of the desired claim.

3. Reduce the dynamics without discarding a state

Probability deviations sum to zero, so the fourth deviation equals minus the first three. This leaves three independent coordinates. A fixed invertible basis changes how those coordinates are represented; it does not change the dynamics. The basis is near a numerical eigenbasis, but the proof does not assume its entries are exact eigenvectors. Its decimal entries and inverse are enclosed directly.

The generator's slow eigenvalue is found inside a bracket where the characteristic polynomial has opposite endpoint signs and a positive derivative. Interval Newton steps narrow the bracket while retaining every possible root. A separate bound proves that both other decay rates are faster. This justifies using the slow relaxation time $\tau = -1/\lambda$, including how that clock changes with the uncertain physical parameters.

4. Bound the matrix exponential and its truncation error

The slow part evolves exactly as e^{-u} times its spectral projector, where $u = t/\tau$. The remaining two-dimensional block is enclosed by splitting it into a diagonal part and an off-diagonal part. The first retained terms can be evaluated explicitly. The omitted terms are bounded by a convergent exponential-series tail; their parameter derivatives receive their own tail bounds.

A second formula expresses the same fast-block exponential using hyperbolic functions. It evaluates the entire functions $\cosh(\sqrt{z})$ and $\sinh(\sqrt{z})/\sqrt{z}$ through formulas that remain finite at $z = 0$. This avoids dividing by a fast eigenvalue gap that might vanish. For the relevant obligations, the checker intersects this enclosure with the truncated-series enclosure.

"The neglected terms look small" would be insufficient. The actual proof adds an explicit bound for all neglected terms. Tests near zero and on both sides of the implementation's series/formula boundary check that the two evaluation branches behave consistently.

5. Accept cells only when the complete inequality follows

For the distance benchmark, each accepted cell must establish both $0.9D_{\text{cold}} - D_{\text{hot}} \geq 0$ and $D_{\text{cold}} \geq 0.001$. The second condition prevents an apparently impressive relative advantage from being certified solely at a distance below the declared floor.

If an interval sign is unresolved, divide a coordinate interval at a shared midpoint. The two closed children cover their parent exactly. Repeat until each leaf is accepted. A heuristic may choose the split coordinate, but the heuristic never supplies the sign proof.

Success requires an empty queue. A node limit with pending cells means incomplete coverage. It does not prove that the physical claim is false, and it cannot be presented as a complete certificate. A deliberate zero-node-budget test confirms that this implementation returns an incomplete result with one pending cell.

6. Keep each layer of evidence in its proper role

The independent checker uses the original four-state matrix and 70-digit direct exponentials, avoiding the reduced spectral implementation. Its 6,744 value comparisons and 360 derivative comparisons passed. The primitive boundary suite adds 295 comparisons and invalid-domain checks. Equal-preparation controls correctly reject false distance and time-saving claims.

Those checks challenge implementation errors. The continuous conclusion additionally needs the analytic enclosure argument and completed whole-domain coverage. The 54-step review records that chain of reasoning.

▼ **Why eight proof obligations cover all temperatures, times, and thresholds**

1. Specify what the uncertainty actually changes

The box perturbs eight independent coordinates: three energy gaps, three shared barrier excesses, and two temperature excesses $T/T_{bath}-1$. Each ranges over ±5% of its nominal value. The rates are rebuilt from these quantities, preserving detailed balance. This is a different domain from independently changing eight matrix entries or changing the full temperatures by ±5%.

2. Put every threshold crossing inside one common time interval

The threshold band is $0.001 \le \delta \le 0.01$. Three endpoint obligations establish that both preparations have KL distance above 0.01 at $u = 0.1$ and below 0.001 at $u = 2$. KL decreases strictly along each nonequilibrium trajectory of this positive reversible model, so each threshold has exactly one crossing in that time interval.

This monotonicity is a mathematical property of the trajectory, not an inference from a plot. In density ratios $r_i = p_i/q_i$, the time derivative is a negative sum of products $(r_i - r_j)(\ln r_i - \ln r_j)$. Every product is nonnegative because logarithm is increasing; positivity of all transition rates gives strictness away from equilibrium.

3. Show that the earliest cold crossing is controlled by an endpoint temperature

A continuous derivative obligation proves that cold KL is concave as a function of its preparation temperature for all relevant physical parameters and times. A concave graph lies above the chord connecting its endpoints.

Fix a threshold and take the earlier of the two cold endpoint crossing times. At that time both endpoint distances are at least the threshold. The chord is therefore at least the threshold, and so is every interior cold temperature's distance. Every interior cold crossing must occur at least as late as that earlier endpoint crossing.

4. Show which hot temperature is worst

Another obligation proves $\partial D_{hot}/\partial T \le 0$ whenever its distance is in the threshold band. At the crossing, implicit differentiation gives

$$\partial u_{hot}/\partial T = -(\partial D_{hot}/\partial T)/(\partial D_{hot}/\partial u) \le 0.$$

The denominator is negative. Thus increasing the hot preparation temperature cannot delay its threshold crossing in the certified region. The lowest allowed hot temperature is the worst case for this comparison.

5. Prove two boundary-temperature implications

It remains to compare that lowest hot temperature against each cold endpoint. Two obligations prove, throughout the six-dimensional physical box and the relevant time interval,

$$0.001 \le D_{cold}(u) \le 0.01$$
$$\Rightarrow D_{hot}(0.95u) \le D_{cold}(u).$$

At a cold threshold crossing, this means the hot trajectory has reached the same threshold by at most 95% of the cold time. The preceding temperature arguments then extend those two comparisons to every allowed pair of preparation temperatures.

6. Use the threshold condition without discarding difficult cells

Let F = $D_{cold}(u) - D_{hot}(0.95u)$. On the threshold band both $D_{cold} - 0.01$ and $0.001 - D_{cold}$ are nonpositive. Therefore, for any chosen $\alpha \geq 0$, either of these whole-cell inequalities is sufficient:

$$F + \alpha(D_{cold} - 0.01) \geq 0,$$
$$F + \alpha(0.001 - D_{cold}) \geq 0.$$

The multiplier can reduce interval overestimation. Its proposed value is a heuristic, but acceptance requires a rigorous bound on the resulting expression. A cell may be excluded only when its entire distance enclosure lies outside the band. Moving a minimum to a face does not permit excluding the original cell merely because that face lies outside the band.

7. Count the obligations and bind them to the exact inputs

The eight obligations are: one distance window; both early endpoints; cold late endpoint; cold temperature concavity; hot temperature monotonicity on the band; hot late endpoint; and two boundary-temperature time comparisons. All are needed for the stated combined conclusion.

The runner checks that the source's embedded decimals exactly equal the requested JSON decimals. This matters because 1.00000000000000001 and 1.00000000000000002 convert to the same ordinary double. The former float-only comparison could miss such a mismatch. The corrected comparison rejects it before compilation; the current model inputs already matched exactly, and the separate completion audit had also checked exact equality.

See the analytic reduction, coverage runner, input regressions, and negative controls. These certify mathematical claims about the specified idealized rate model; laboratory realization and preparation costs require separate evidence.

The following eight walkthroughs reproduce Lessons 33–40 of the expanded report. Equations use native MathML and work offline.

▼ **Lesson 33. A rate is not a probability: build the clock first**

Begin with units.

A probability is a number between zero and one. A rate has units of inverse time. A rate of three events per second is perfectly meaningful; a probability of three is not. If an event has constant rate r, its probability of occurring in a very short interval of length h is $rh + o(h)$. The notation $o(h)$ means that the

error divided by h tends to zero as h tends to zero. It does not mean that the approximation is exact for a one-second interval.

Derive the waiting law.

Let $S(t)$ be the probability that no event has happened by time t. Assume that, conditional on no event so far, the chance in the next short interval is still $rh + o(h)$. Then

$$S(t+h) = S(t)(1 - rh + o(h)), \qquad S'(t) = -rS(t), \qquad S(0) = 1.$$

The function with this derivative and initial value is $S(t) = e^{-rt}$. Thus the probability of at least one event by time t is $1 - e^{-rt}$. For $r = 2$ per second and $t = 0.1$ seconds, this is $1 - e^{-0.2}$, approximately 0.1813. The short-time estimate $rt = 0.2$ is close but different. For $t = 1$ second the estimate $rt = 2$ is not a probability at all.

Why the mean wait is $1/r$.

The density of the first event time is re^{-rt}. Its mean is

$$\int_0^\infty tre^{-rt}\,dt = [-te^{-rt}]_0^\infty + \int_0^\infty e^{-rt}\,dt = \frac{1}{r}.$$

The integration by parts has merely moved the factor t onto a derivative. There is also a geometric interpretation: the mean waiting time is the area under the survival curve $S(t)$.

Several possible exits.

From state j, let the exit rates to other states be a_{ij}. Independent exponential clocks compete. The probability that none has rung by t is the product $\prod_{i \neq j} e^{-a_{ij}t}$, hence the total exit rate is $r_j = \sum_{i \neq j} a_{ij}$. Conditional on an exit, its destination is i with probability a_{ij}/r_j. This produces the diagonal entry $M_{jj} = -r_j$ and explains why a column sums to zero. The negative diagonal does not describe a negative probability; it describes probability leaving.

Try it.

Suppose the rates from one state to two others are 2 and 3 per second. The mean residence time is $1/5$ second, and the next destination probabilities are $2/5$ and $3/5$. Multiplying all rates by ten divides every waiting time by ten. It does not change these destination probabilities or the equilibrium distribution. Consequently, a percentage time advantage alone supplies no absolute number of seconds without a physical rate calibration.

▼ **Lesson 34. Replace unequal clocks by one clock and harmless pauses**

A deliberately simple two-state example.

Use a column probability vector and let

$$M = \begin{pmatrix} -2 & 1 \\ 2 & -1 \end{pmatrix}, \qquad \rho = 2, \qquad P = I + M/\rho = \begin{pmatrix} 0 & 1/2 \\ 1 & 1/2 \end{pmatrix}.$$

The first state exits at rate 2; the second exits at rate 1. Instead ring a clock of rate 2 in either state. At each ring, choose the next state using the current column of P. In state two, half the rings leave the state unchanged. Its actual transition rate is therefore $2 \times (1/2) = 1$, exactly as required. The pauses are real elapsed time.

Generalize only after the example.

Choose ρ at least as large as every exit rate. Then $P = I + M/\rho$ has nonnegative entries and unit column sums. Such a matrix is called stochastic: it sends probability vectors to probability vectors. The identity $M = \rho(P - I)$ gives

$$e^{Mt} = e^{\rho t(P-I)} = e^{-\rho t} \sum_{k=0}^{\infty} \frac{(\rho t)^k}{k!} P^k.$$

The factorization is allowed because P commutes with I. It would not be legitimate to split an arbitrary matrix exponential into two factors without checking commutation.

Read every factor as a probability.

Write $\mu = \rho t$. A Poisson random variable N has probabilities $\Pr(N = k) = e^{-\mu}\mu^k/k!$. Here N counts clock rings by time t. Conditional on $N = k$, the distribution is $P^k p(0)$. Averaging these conditional distributions gives the displayed formula. This procedure is called uniformization. It is an exact continuous-time construction, not a small-time-step approximation.

At $\mu = 1$, the probabilities of zero, one and two rings are respectively e^{-1}, e^{-1} and $e^{-1}/2$. Their sum is about 0.9197. The missing 0.0803 is the probability of three or more rings; it cannot simply be discarded when a rigorous error bound is required.

Check the two-state equilibrium.

Solving $M\pi = 0$ and $\pi_1 + \pi_2 = 1$ gives $\pi = (1/3, 2/3)^{\mathsf{T}}$. Starting from $(1, 0)^{\mathsf{T}}$, the exact first component is $p_1(t) = 1/3 + (2/3)e^{-3t}$ and the second is its complement. Differentiating verifies $p_1' = 1 - 3p_1$, which is also

the first row of Mp. This small solvable model is a useful check of the matrix convention, the clock interpretation and any implementation of uniformization.

▼ **Lesson 35. Bound the part of an infinite sum that you did not compute**

First learn the geometric comparison.

For $0 \le q < 1$, multiplying $1 + q + q^2 + \cdots$ by $1 - q$ cancels every term except the first, giving $1/(1-q)$. If a nonnegative series starts with a and every later term is at most q times its predecessor, its whole sum is at most $a/(1-q)$. Equality is unnecessary; an upper bound suffices.

For Poisson weights $w_k = e^{-\mu}\mu^k/k!$,

$$\frac{w_{k+1}}{w_k} = \frac{\mu}{k+1}.$$

After keeping terms 0 through K, the first omitted weight is w_{K+1}. Every subsequent ratio is at most $q = \mu/(K+2)$. Provided $q < 1$,

$$R_K = \sum_{k=K+1}^{\infty} w_k \le \frac{w_{K+1}}{1 - \mu/(K+2)}.$$

The $K+2$ in the denominator is intentional: it bounds the ratio from the first omitted term to the next omitted term. Confusing it with the first omitted term's index is an easy off-by-one mistake.

Work a numerical example.

Take $\mu = 1$ and keep $K = 2$. Then $w_3 = e^{-1}/6$ and $q = 1/4$, so $R_2 \le 2e^{-1}/9$, approximately 0.08175. The actual missing mass is approximately 0.08030, below the bound. The bound is a little wider than necessary, but its validity is transparent.

Turn a scalar tail into a vector guarantee.

Every component of $P^k p(0)$ lies in $[0, 1]$. If s_i is the computed partial sum for component i, then $p_i(t) \in [s_i, s_i + R_K]$. These independent intervals need not sum to exactly one; they are an enclosure of a probability vector, not themselves a new probability vector. Renormalizing their endpoints without a proof could invalidate the enclosure.

Include arithmetic error as well as truncation error.

An exact tail formula does not excuse rounding errors in the partial sum. The nominal checker computes interval bounds for inputs, exponentials, matrix products and logarithms. If the entire probability interval

is positive, it can enclose each term $p_i \log(p_i/\pi_i)$ and hence KL. An interval crossing zero requires a valid alternative treatment or a failure to certify; a logarithm of an invalid endpoint is not evidence.

In the replayed nominal calculation, at most 281 terms were needed and the largest reported tail upper bound was below 10^{-45}. The small tail is one part of the certificate. The validity of P, the input correspondence, the outward arithmetic and the later comparison inequalities remain separate necessary parts. More printed digits alone establish none of them.

▼ **Lesson 36. Prove a whole time interval using its endpoints**

A descending staircase picture.

Suppose two distances $D_c(t)$ and $D_h(t)$ are nonincreasing in time. On a cell $[a, b]$, the cold distance is at least its value at the right endpoint, while the hot distance is at most its value at the left endpoint:

$$D_c(t) \geq D_c(b), \qquad D_h(t) \leq D_h(a).$$

Therefore a single strict inequality

$$D_h(a) < 0.9 D_c(b)$$

proves $D_h(t) < 0.9 D_c(t)$ for every time in that cell. The endpoints are deliberately mismatched to use the worst case for each curve. Merely checking $D_h(a) < 0.9 D_c(a)$ and the analogous inequality at b would not control a crossing between them.

Why KL decreases here.

For a stochastic transition matrix K and positive equilibrium π, let $r_i = p_i/\pi_i$. When $K\pi = \pi$, the output likelihood ratio is

$$\frac{(Kp)_j}{\pi_j} = \sum_i \frac{K_{ji}\pi_i}{\pi_j} r_i.$$

The coefficients in this sum are nonnegative and sum to one. Convexity of $f(r) = r \log r$ says that f of this weighted average is no greater than the weighted average of f. Multiply by π_j, sum over j, and use $\sum_j K_{ji} = 1$ to obtain

$$D(Kp \| \pi) \leq D(p \| \pi).$$

Apply this to $K = e^{M(b-a)}$. This argument proves the monotonicity used above. It does not say that $D_h(t)/D_c(t)$ is monotone.

A hand-checkable cell.

Suppose outward bounds give $D_h(a) \le 0.0080$ and $D_c(b) \ge 0.0090$. Then $0.9D_c(b) - D_h(a) \ge 0.0001 > 0$. Every interior time passes the ten-percent comparison. If the computed margin interval were $[-0.0001, 0.0002]$, no conclusion would follow; a narrower cell might resolve it.

Join the cells and check the floor.

A finite list $t_0 < t_1 < \cdots < t_N$ covers the entire closed interval $[t_0, t_N]$ without gaps. Proving the endpoint inequality on every cell proves the full window. Since cold KL decreases, a lower bound $D_c(t_N) \ge 10^{-3}$ also proves that floor throughout the window. The nominal replay used 80 cells and a minimum certified comparison margin greater than 0.0004408. Its fixed endpoints were approximately 58.89269 and 117.78539 in attempt-time units. The script separately encloses the slow time to justify the window-length requirement; a decimal time chosen from a numerical eigenvalue does not become exact merely by being described as a multiple of that numerical estimate.

▼ **Lesson 37. Prove a whole band of target distances**

Reverse the graph in your mind.

For a positive target δ below the initial distance, define $t_x(\delta)$ as the first time preparation x reaches that target. A smaller target means getting closer to equilibrium, so it takes longer. Thus if $\delta_a < \delta_b$, then $t_x(\delta_a) \ge t_x(\delta_b)$. For the irreducible reversible model and a nonequilibrium preparation, KL strictly decreases until its limiting equilibrium value; positive targets below the initial value have unique crossings.

A root estimate proposes; a sign bracket verifies.

A numerical solver might propose $t = 10.1234$. To verify a bracket $[L, U]$, enclose $D(L) - \delta$ strictly above zero and $D(U) - \delta$ strictly below zero. Continuity gives a crossing between them; strict decrease gives uniqueness. An approximate residual close to zero is weaker than these two certified signs. We also need $0 < L < U$ for the ratio bound below.

Choose the unfavorable corners.

For every $\delta \in [\delta_a, \delta_b]$,

$$t_h(\delta) \le t_h(\delta_a) \le U_h(\delta_a), \qquad t_c(\delta) \ge t_c(\delta_b) \ge L_c(\delta_b) > 0.$$

Division by a positive denominator preserves the intended order, giving

$$1 - \frac{t_h(\delta)}{t_c(\delta)} \geq 1 - \frac{U_h(\delta_a)}{L_c(\delta_b)}.$$

If the right side exceeds 0.05, every target in that cell has at least five-percent time saving. Notice that the numerator and denominator use opposite ends of the target cell. This conservative mismatch is exactly what covers its interior.

Try the ratio with ordinary numbers.

Suppose the hottest trajectory's slowest permitted arrival is at most 90 seconds and the cold trajectory's fastest permitted arrival is at least 100 seconds. The saving is at least $1 - 90/100 = 10\%$. If the denominator were instead only known to exceed 92, the bound would fall to about 2.17%, even if all sampled same-target comparisons looked better. The proof must use the bounds it has actually established.

The nominal certificate covers $[10^{-3}, 10^{-2}]$ with 60 target cells. Its conservative lower bound on time saving is approximately 8.12245%. This is a guaranteed lower bound for that nominal model, not the exact minimum. The later eight-obligation certificate handles the full uncertain four-state parameter box. It is a distinct proof, not a conclusion obtained by silently extrapolating the nominal calculation.

▼ **Lesson 38. Derive a transition rate from an energy exchange**

Begin with one uphill move.

Let two system energies be $E_j = 1$ and $E_i = 3$, with barrier energy $B = 5$. Draw a nonnegative reservoir energy Y with density $\beta e^{-\beta Y}$. The move from j is allowed when $1 + Y \geq 5$, namely when $Y \geq 4$. At acceptance, set $Y' = Y + 1 - 3 = Y - 2$. It is nonnegative, and $E_i + Y' = 3 + Y - 2 = 1 + Y = E_j + Y$: combined energy is conserved. The reverse contact sees the same total energy and can undo the move.

Integrate the acceptance probability.

For any $c \geq 0$,

$$\Pr(Y \geq c) = \int_c^\infty \beta e^{-\beta y}\, dy = e^{-\beta c}.$$

If contacts for this pair arrive at rate ν_0, the accepted rate from j to i is therefore

$$a_{ij} = \nu_0 e^{-\beta(B - E_j)}.$$

The reverse rate is $a_{ji} = \nu_0 e^{-\beta(B - E_i)}$. Their ratio is $e^{-\beta(E_i - E_j)}$. Multiplying by Gibbs weights gives the stronger flux identity

$$a_{ij}e^{-\beta E_j} = a_{ji}e^{-\beta E_i} = \nu_0 e^{-\beta B},$$

which is detailed balance before the common normalization is inserted.

Separate an always-required gate from the uphill cost.

Write $h = B - \max(E_i, E_j)$. Since $B - E_j = h + \max(E_i - E_j, 0)$, the rate becomes

$$a_{ij} = \nu_0 e^{-\beta h} \min\{1, e^{-\beta(E_i - E_j)}\}.$$

In the example at $\beta = 1$, the common gate contributes e^{-2}, the uphill acceptance contributes another e^{-2}, and their product is e^{-4}. The reverse move has only the common e^{-2} gate.

Why simulation may omit attempts but not their time.

One can retain only contacts with $Y \geq h$. Their clock has rate $\nu_0 e^{-\beta h}$. Conditional on passing the gate,

$$\Pr(Y - h \geq z \mid Y \geq h) = \frac{e^{-\beta(h+z)}}{e^{-\beta h}} = e^{-\beta z}.$$

Thus a retained mark is h plus a fresh exponential energy. This proves the thinning rule used in the event simulation. Counting every attempted contact as an accepted transition would alter the process; ignoring rejected contacts without adjusting the clock would alter physical time.

What this construction establishes.

Under independent fresh thermal marks, independent Poisson pair clocks, constant reservoir density of energy states, and instantaneous updates, the rate identity holds exactly for every allowed barrier and energy. It supplies a stochastic bath interpretation of the same four-state model. It does not identify a material or derive an autonomous microscopic bath. The replayed simulation of 500,000 trajectories per preparation agrees with direct evolution at five times to within 0.00118 per component. That is a diagnostic check of the simulation; the rate identity above is the reason the mathematical generators coincide.

▼ **Lesson 39. An intermediate state introduces a second clock**

Insert a waiting room between two wells.

Suppose the desired well-to-well rates are u from j to i and v from i to j. Insert state s. Use entrance rates $2u$ from j and $2v$ from i, and exit rate b from s to each well. On entering s, the next well is selected with probabilities $1/2$ and $1/2$. Half the visits return to the well just left.

If the clock is stopped during every visit to s, the successful transition rate from j to i is $2u \times (1/2) = u$. This is the trace clock: only time actually spent in the wells is retained. With ordinary physical time, each visit adds a mean wait $1/(2b)$. Those waits cannot be removed when claiming a physical completion time.

Derive the block identity one entry at a time.

Order the wells as (j, i). Their direct block, the entrance block, the exit block and the intermediate block are

$$Q_{ww} = \begin{pmatrix} -2u & 0 \\ 0 & -2v \end{pmatrix}, \quad Q_{sw} = \begin{pmatrix} 2u & 2v \end{pmatrix}, \quad Q_{ws} = \begin{pmatrix} b \\ b \end{pmatrix}, \quad Q_{ss} = -2b.$$

The subscripts give destination first, as everywhere else. Multiplication now gives

$$Q_{ww} - Q_{ws}Q_{ss}^{-1}Q_{sw} = \begin{pmatrix} -2u & 0 \\ 0 & -2v \end{pmatrix} + \begin{pmatrix} u & v \\ u & v \end{pmatrix} = \begin{pmatrix} -u & v \\ u & -v \end{pmatrix}.$$

This is exactly the intended two-well generator. Adding the six pair contributions gives the four-well identity in the technical report. This exact algebra identifies the trace generator; it does not assert that the physical-time well probabilities solve that same four-state equation.

Equilibrium and preparation also change.

In the expanded model, each intermediate has energy B and phase-space weight $\eta > 0$. Its equilibrium weight is ηe^{-B}, compared with e^{-E_j} for a well. Taking $b = 2/\eta$ in attempt-time units gives mean intermediate residence time $\eta/4$ and matched equilibrium fluxes. At preparation temperature T, all ten states must be prepared with weights $g_a e^{-E_a/T}$. Preparing only the wells would be a different experiment. The phase-space weight η need not be an integer count.

At $\eta = 0.05$, the nominal cold and hot preparations place about 1.90% and 4.07% of their mass in intermediates. Full ten-state calculations retain those populations and every residence time. Their finite-time advantage persists at the tested points. The expanded system also has higher energies and faster rates outside the original all-state caps. It is a separate test of a modeling assumption, not a new certificate inside the original constrained four-state class.

▼ **Lesson 40. Read a computational result at the right strength**

Four questions before accepting a result.

What object was calculated? What inputs were used? Which errors were bounded? Which variables were covered continuously? An answer such as “70 digits” only addresses numerical precision, and even then does not itself establish a bound on every implementation or modeling error.

The independent ten-state calculation directly constructs and exponentiates the full generator at 70-digit working precision. It agrees with the fast spectral implementation within 6.55×10^{-13} in KL at five tested times. Different formulas agreeing is useful evidence against a numerical implementation mistake. Neither implementation uses outward error bounds for that comparison, so it is a high-precision diagnostic, not an interval certificate.

Interpret a large sample precisely.

For each of three fixed η values, the study tests 256 corners and 500 seeded interior parameter points. The total is $3(256 + 500) = 2268$. Each point is further tested on finite target and time grids. No sampled case fails the declared benchmarks. This says exactly what happened on those grids. It does not exclude a narrow failure between points, between time samples, between targets, or at an untested value of η. The continuous four-state certificate cannot be transferred to this ten-state extension merely because their trace generators agree.

A percentage needs a denominator and a clock.

At $\eta = 0.05$ and target 10^{-3}, the independently computed arrival times are approximately 210.67249 for cold and 190.05131 for hot. Using the cold time as denominator gives

$$\frac{210.67249 - 190.05131}{210.67249} \approx 0.0978826,$$

or about 9.78826%. Dividing by the hot time would answer a different percentage question. These are attempt-time units, not calibrated seconds. The calculation includes post-quench intermediate residence times but does not include the cost or duration of preparing the hotter initial state.

Know what a program's success message establishes.

A zero exit code means that the process completed without a reported execution failure. Some exploratory programs record failures in JSON without returning a nonzero code. Read the recorded failure counts, input hashes, unresolved cells, parameter scope and numerical residuals. A certificate runner must also reject incomplete coverage. A numerical search that found nothing has a different meaning from a complete proof that nothing exists.

The final handoff to advanced mathematics.

You can now read the physical section by following a chain of explicit claims: construct valid rates; establish equilibrium; evolve preparations; bound numerical error; prove continuous comparisons; and interpret the clock and physical assumptions. The advanced spectral and interval methods make that chain efficient over an uncertain parameter box. They do not replace any link. When a later formula looks forbidding, identify which link it serves before trying to manipulate its symbols.

▼ **Why approaching 25% can require enormous energy separation**

1. A limit does not tell you how quickly you get there

The sequence 1–1/n and the sequence 1–1/log(n) both approach 1 as n grows, but the second approaches much more slowly. In the report, the amplitude advantage approaches 25%. A separate calculation is needed to describe the remaining gap.

2. Replace a tiny parameter with a large, readable one

The family uses a small positive number ε. Set L = log(1/ε), so ε = exp(−L). Its highest energy is 2L in bath-scaled units. Increasing L means increasing the energy separation. Define a = log(5/3), y = exp(−aβ), and x = exp(−2Lβ). The third-state weight y changes slowly with inverse temperature β; the highest-state weight x changes quickly.

```
Leading positive slow amplitude:
R(β) = (y/3 − 1/5)/(2 + y + x).
```

The exact formula differs by order ε on the cooling interval. “Order ε” means bounded by a constant times ε for sufficiently small ε, not equal to ε. Differentiation can introduce the large factor 2L, so the derivative error must be checked separately; it is order Lε.

3. Locate the balance between two competing changes

Emptying the highest state reduces the denominator and initially increases the amplitude. Changing the third-state weight eventually reduces the numerator enough to reverse that trend. Differentiate the quotient, using y′ = −ay and x′ = −2Lx. After multiplying by the positive denominator factor, the leading stationary-point equation is:

```
x[(10L−5a)y−6L] − 13ay = 0.
```

Near the maximum, y approaches 1. The bracket therefore behaves like 4L, giving x approximately 13a/(4L). Since x = exp(−2Lβ), taking logarithms gives the first approximation to the maximizing inverse temperature:

$$\beta_m \approx \log(4L/(13a))/(2L).$$

The report also proves uniqueness using its earlier bound on stationary points. A numerical root solver by itself cannot establish that there are no other maxima.

4. Control the error in stages

The initial equation still contains an error proportional to β_m. First absorb that term into the left side to prove β_m is at most a constant times log(L)/L. Only then substitute the new bound back to obtain the smaller error of order log(L)/L². This two-stage argument prevents using an unproved estimate to justify itself.

$$\text{Advantage} \approx 1/4 - [13a/(16L)]\cdot[\log(4L/(13a)) + 1].$$

The remaining deficit is of order log(L)/L. Because the highest energy is 2L, this explains why a clear separation between slow and fast decay modes does not guarantee a near-25% advantage at modest energy separation.

5. Compare the same quantities

This formula compares the peak with the infinite-temperature limit β = 0. The figure instead uses a finite hot preparation β = 1/L². Both approach 25%, but they have different smaller correction terms. The graph must not be treated as a direct plot of the displayed asymptotic formula.

The independent checker verifies six exact algebra identities and recomputes five plotted maxima. Its numerical residuals illustrate the predicted error scales; the analytic derivation proves them.

▼ **What the cited physical literature establishes**

Metropolis and colleagues (1953) supply the familiar energy-dependent acceptance rule. The model's Poisson clock needs its own justification.

Creutz (1983), pages 1411–1412, describes both a retained energy-carrying demon and thermal refreshment. Fresh auxiliary energy is therefore established prior art.

Biswas, Rajesh and Pal (2023), Sections IV and VI, explain population mechanisms and show that metastability is not necessary. Their paper does not certify this project's rates or uncertainty box.

These are checked source correspondences, not an exhaustive priority search. The review record identifies the inspected locations and the limits of each attribution.

▼ **Read a figure without confusing its model, units, or evidence**

1. Name the experiment before reading the curve

Several figures appear in this project, but they do not all describe the same system. The primary counterexample and the rational companion share their energy weights but have different rates. The controlled mechanism changes a mode parameter. The quarter-limit family changes its energy separation. The Arrhenius study is a separate physical model.

Figure	What varies?	What is measured?
Primary thermal overlap	Preparation inverse temperature	Absolute slow-mode amplitude
Primary relaxation	Elapsed time under one fixed generator	KL distance and the hot/cold ratio
Controlled onset	Mode component z and preparation	The location of the amplitude maximum
Quarter-limit family	Energy separation and preparation	Amplitude advantage with finite hot temperature
Physical-model study	Time or a common KL target	KL distance or percentage of time saved
Focused overview, panel A	Time for the rational companion	A quadratic distance, D_q

2. Check the denominator in a percentage

If the hot amplitude is 99 and the cold amplitude is 100, the reduction relative to cold is 1%. The increase relative to hot is 100/99−1, or approximately 1.0101%. Both calculations are correct, but they answer

different questions. The primary overlap enlargement uses the hot reference; the stated amplitude advantage uses the cold reference.

A relative-distance ordinate $100(D_h/D_c-1)$ becomes negative when the hot preparation is closer. A time-saving ordinate $100(1-t_h/t_c)$ becomes positive when the hot preparation arrives earlier. Always read the actual formula before interpreting the sign.

3. Check the clock and distance normalization

The primary and rational examples measure time in units set by a particular transition rate. The physical figure uses t/τ, where τ is its own slow relaxation time. Equal horizontal coordinates on these plots do not mean equal seconds.

The rational overview uses $D_q = \Sigma(p_i-\pi_i)^2/q_i$ with unnormalized weights q. Standard Pearson chi-square instead divides by $\pi_i = q_i/Q$, so it equals $Q\cdot D_q$. The multiplier cancels in a hot/cold ratio for this fixed model, but the two raw distance values differ.

4. A shaded region needs a definition

The physical time-window shading identifies the interval covered by the separate certificate. The threshold-panel shading instead shows the lowest and highest values observed in a finite parameter sample. Connecting sampled extrema does not enclose all unsampled parameters or thresholds. The certified 5% floor comes from the interval proof, not from the shaded envelope.

5. Reproduce first, then cross-check independently

The figure audit rebuilt all five plotting programs in an isolated directory. Their CSVs and PNGs matched the saved outputs exactly; PDF differences were confined to creation dates. A second checker then used direct matrix exponentials and independently differentiated thermal sums to check the numerical data.

Across all nine figure CSV files, the independent checker passed 6,238 comparisons. The separate 3,256-point physical parameter scan also reproduced its complete saved result. These checks establish numerical correspondence. The exact and interval certificates remain the evidence for the mathematical claims.

Inspect the comparison errors, isolated reproduction receipt, and formula and caption review.

▼ **From trying examples to proving a whole claim: a slow introduction**

1. Start with one adjustable number

Suppose you can choose a number x between 0 and 1. For each choice, a calculation gives a score f(x). You want to know whether the score stays positive. Trying x = 0 and x = 1 gives two answers. It does not yet tell you what happens between them.

Take $f(x) = (x-½)^2-1/10$. At either endpoint, $f = ¼-1/10 = 3/20 = 0.15$. At the middle, $f(½) = -0.1$. Both endpoint tests pass, while an interior test fails. The issue is not that the computer calculated incorrectly. The tested points did not cover the claim.

2. Add a second adjustable number

Two parameters make a rectangle of possible choices. Checking its four corners still says nothing about an arbitrary nonlinear function inside it. With eight parameters there are $2^8 = 256$ corners, but infinitely many interior points. A corner reduction requires an additional mathematical argument, such as suitable monotonicity in every parameter. Counting corners is not that argument.

Checking one parameter at a time is weaker still. For $f(x,y) = 0.1-xy$ with x and y between −1 and 1, each coordinate axis has f = 0.1. Changing both parameters to 1 gives f = −0.9. Interactions can hide entirely from single-parameter tests.

3. Translate a cooling target into a numerical score

Fix a target distance δ from equilibrium. Let t_c be the time the cold preparation first reaches that target. A 5% time saving means the hot preparation reaches it by $0.95t_c$. For the decreasing KL distance in this study, it is enough to check $D_h(0.95t_c) \le \delta$.

Divide by the positive number δ and subtract 1. The resulting score is $D_h(0.95t_c)/\delta-1$. A score at most zero passes the target. If there are several tested thresholds, take the largest score: every tested threshold passes exactly when that maximum is at most zero. This still covers only the thresholds and parameters actually tested.

4. Understand what an optimizer does

An optimizer chooses new parameter values using previous scores. A random seed makes its random choices reproducible under suitable software conditions. It does not turn a finite search into a proof. Different library versions or floating-point arithmetic can change which point is tried next.

Two questions must remain separate: “What was the best point found?” and “Did the algorithm satisfy its stopping rule?” A run can stop at its iteration limit, report `success=false`, and still return a useful point. All three saved adversarial searches here have that status. Their candidate savings can be checked directly; their status does not establish that a better or worse point is impossible.

5. Learn from the rejected candidates

The first physical search optimized the nominal system. The second screened individual parameter changes. The refinement then evaluated combinations at all 256 corners and used adverse corners to guide further searches. Its first accepted improvement still had a positive worst-corner score, about 0.03630: it was better than its predecessor, but still failed a target. The next stages made that tested worst score negative, about −0.00365.

“Accepted by a refinement algorithm” therefore means “better under that algorithm's comparison,” not “proved to satisfy the research goal.” The eventual continuous guarantee comes from the separate interval certificates and their analytic reductions.

6. Check the evidence without inheriting the search's calculation

The new audit recomputed all 153 saved improving search points, both final search scores, the refinement scores and decisions, and the three adversarial candidate savings. It used direct matrix exponentials in physical time rather than the search's spectral propagation. All 166 numerical comparisons passed; the largest absolute discrepancy was below 8×10^{-13}.

That agreement checks the recorded point values. It does not, by itself, authenticate historical elapsed times, every unsaved trial, or a global optimum. A fresh replay and a mathematical certificate answer different questions, and both should retain their stated scope.

Inspect the independent calculation and its source-bound receipt. The exact rational examples above require only multiplication and subtraction; the advanced physical certificate uses interval bounds to cover unsampled points.

▼ **Why a promising candidate can fail: from one inequality to exact rejection**

1. A counterexample must satisfy every assumption

Suppose a claim says, “Every number x satisfying $x \le 0$ also has property P.” A proposed counterexample needs both $x \le 0$ and failure of P. The number $x = 0.000000001$ does not qualify, even if it is very close to zero. Replacing the condition by $x \le 0.000001$ changes the problem.

A numerical solver may allow a small residual when it tests an inequality. That tolerance is useful during exploration, but the exact mathematical claim keeps its original boundary. In one retained inverse search,

a numerical candidate was promising enough to save. Exact rational reconstruction then found seven positive triplet tests, including $F_{234} \approx 1.30\times10^{-7}$. It was correctly rejected.

2. A prescribed mode can be correct while the rates are wrong

The first rational backward design had the desired eigenvector and eigenvalue. That means its linear eigen-equation was satisfied. Nevertheless, its 1–2–4 triplet had positive F and G, so it violated the required premise. Solving the eigen-equation was only one part of the job. A revised choice of adjustable rates satisfied the premise and matches the independently certified rational companion.

3. A ratio can magnify a nearly zero error

Consider a diagnostic ratio a/b. If both a and b should be almost zero, tiny rounding errors can dominate the ratio. For example, a computed numerator -10^{-16} divided by a computed denominator 10^{-16} gives –1. A large-looking ratio need not represent a large physical or mathematical violation.

A saved six-state search point produced a negative arithmetic diagnostic. Its high-precision recheck did not confirm a violation. We can now reject it exactly: subtracting the minimum reset leaves two connected groups, states {1,2,3,4} and {5,6}. The relevant slow mode is constant within each group. Calling its first-group value a, the tested expression is $w_1+w_2-2w_3 = a+a-2a = 0$.

The advanced justification uses reversibility: the internal dynamics has exactly two zero modes, one per connected group, while all its other modes decay. Adding the reset leaves one equilibrium mode and shifts the other component-constant mode to eigenvalue –1. Every remaining mode is faster. Thus the zero arithmetic margin belongs to the actual simple slow mode. This settles this stored candidate, not the general higher-dimensional question.

4. Count attempts and retained checks separately

The broad heating search recorded about 6.74 billion attempted derivative evaluations. Positivity and signal-size filters excluded some points from the score comparison; spectral and bath-weight filters also restricted which systems were sampled. The later optimizer's roughly 66.4 million objective calls included penalty returns for excluded inputs. Those totals are not counts of certified mathematical cases.

The surviving best scores remained positive, including under separate high-precision calculations. The report now states the filters explicitly. A finite search with no accepted negative score still leaves the general mathematical question open.

5. Keep the successful and unsuccessful records together

The original cone, positive-rate, and signed-rate searches reproduced their saved candidate or final records exactly. The rejected rational design, the seven-inequality failure, and the borderline six-state

case remain available alongside the successful certificates. This makes the reasoning inspectable: a candidate becomes evidence only after the checks appropriate to its claim.

See the exact and high-precision recheck, its receipt, the historical replay comparison, and the bounded report clarification.

▼ **Reproduce a result without confusing the command, its output, and its proof**

1. A command is a recipe with ingredients

A program name alone does not identify a calculation. Its arguments choose inputs and settings; its working directory determines where relative paths begin. For example, `--samples 3000` requests 3,000 interior samples in addition to the 256 corners. Omitting it invokes a different default. A reproducibility record therefore keeps the full command, the inputs, the source version, and the result.

2. A fingerprint identifies a file, not a theorem

A SHA-256 hash is a compact fingerprint of a file's bytes. If an input changes, its fingerprint almost certainly changes. Matching fingerprints help establish that two runs used the same data or code. They do not establish that the code implements the right mathematics. That requires inspecting formulas, assumptions, boundary cases, and error handling.

3. Read the success conditions carefully

A process exit code of zero conventionally means that the program finished successfully. It means only what that program actually checked. The replay wrapper also writes `complete` and `all_passed`: the first says the sequence ended, while the second says every child check passed. Read both. For a continuous interval proof, also check that no cells remain pending and that no enclosure errors occurred.

A useful negative control deliberately supplies a false claim or invalid input. The test passes when the verifier rejects it. This helps check that “success” is not an unconditional printed message. It supplements the proof of the verification method; it does not replace that proof.

4. Separate document rebuilding from checking mathematics

The isolated document rebuild recreated all three LaTeX sources byte for byte. The rebuilt PDFs also had identical page text and drawing content: 126 pages for the report, 81 for the appendices, and 10 for the focused paper. This checks that the advertised recipe preserves the reviewed documents. It does not independently re-prove their theorems.

PDF files can contain creation timestamps, so two files can display exactly the same pages while their whole-file hashes differ. Compare the relevant content before deciding what a difference means.

Preserve the original artifacts while investigating.

5. Understand the formal check's boundary

The Lean command builds the pinned project and checks axiom reports for 43 named theorems. The fresh run passed with only the permitted standard logical axioms. Its mathematical scope is the specific formal statements: it does not automatically cover every numerical figure, physical-model claim, or literature attribution in the report. The guide's earlier lessons explain those separate evidence requirements.

Inspect the command-by-command record, document comparison, wrapper review, and formal check receipt.

STEP 19 · REPRODUCE

Start with a small check you can inspect

Open a terminal in the project root. A terminal runs the text commands below. The first uses only Python's standard library. It imports no project certificate and accepts no floating-point approximation as proof.

```
python3 -I -S research/check_freshman_walkthrough.py
```

Expected result: `PASS: 70 exact checks`, followed by the receipt path. Read the code next to Steps 4–11: it builds the rates, checks signs, reconstructs the characteristic polynomial, encloses fractional powers, and compares intervals.

Check the other witness and formal library

```
python3 -I -S research/standalone_primary_check.py
python3 -I -S research/test_standalone_primary_check.py
./formal/check.sh
```

The first two concern the comprehensive report's primary witness. The third builds the Lean library and checks the dependencies of 43 named theorems. It needs the pinned toolchain and mathlib; see setup and exact coverage. Expected final line: `PASS: 43 theorem axiom audits; standard logical axioms only.`

Replay the wider computations

```
python3 research/recheck_all.py --group core
python3 research/recheck_all.py --group extensions
python3 research/recheck_all.py --group physical
```

These write timestamped logs under `research/detailed_audit/`. The wider groups need the scientific Python packages listed in RUNME. The physical group compiles C++ and needs MPFR/GMP. Use ordinary Python without `-O` for older certificate scripts, because some rely on assertions. The new walkthrough checker uses explicit checks that remain active under optimization.

A script exit code of zero means that run passed its implemented checks. It does not automatically verify every claim in a expanded report. Inspect the coverage, input data, exact tests, and analytic dependencies.

STEP 20 · EVIDENCE AND OPEN WORK

Keep the audit honest and inspectable

Current conclusion: the current report and guide have completed their recorded review. All 68 numbered statement instances have explicit reviews; all 85 advertised command lines are accounted for; all eight continuous physical-box obligations completed with no pending cells or enclosure errors. Analytic reviews and execution receipts are recorded separately in the completion evidence index. Historical exploratory programs are retained as provenance and are not certified as alternative proof implementations.

Exact arithmetic

70 checks passed

Inspect worked-example receipt

Lean build and audit

43 theorem audits passed

Inspect full Lean log

Snapshot generated: 2026-10-13T03:54:35.740610+00:00. Rebuild this page with `python3 research/build_freshman_guide.py` to incorporate new replay receipts.

The test transcription and the finite-temperature scope were checked against the original article, Eq. (3) and its N-level protocol. This is a fixed-version correspondence check, not a claim of exhaustive literature priority.

Fresh replay record

Script	Run status	Evidence
research/standalone_primary_check.py	passed	Run log
research/test_standalone_primary_check.py	passed	Run log
research/adversarial_certificate.py	passed	Run log
research/adversarial_rational_mode_certificate.py	passed	Run log
research/independent_mode_audit.py	passed	Run log
research/finite_time_certificate.py	passed	Run log
research/mechanism_certificate.py	passed	Run log
research/full_triplet_amplitude_certificate.py	passed	Run log
research/extended_arithmetic_certificate.py	passed	Run log
research/extended_mode_obstruction_certificate.py	passed	Run log
research/quarter_limit_certificate.py	passed	Run log
research/quarter_limit_thermal_certificate.py	passed	Run log
research/run_robust_finite_time_certificate.py	passed	Run log
research/ordered_conductance_certificate.py	passed	Run log
research/energy_threshold_certificate.py	passed	Run log
research/classification_exact_certificate.py	passed	Run log

Script	Run status	Evidence
research/degenerate_exact_certificate.py	passed	Run log
research/inverse_realizability_audit.py	passed	Run log
research/test_exact_commensurate_classifier.py	passed	Run log
research/transient_only_chi_certificate.py	passed	Run log
research/five_state_extension_certificate.py	passed	Run log
research/one_fast_mode_certificate.py	passed	Run log
research/compact_endpoint_equality_certificate.py	passed	Run log
research/independent_rational_classifier_audit.py	passed	Run log
research/independent_classifier_nearzero_audit.py	passed	Run log
research/high_dimension_amplitude_certificate.py	passed	Run log
research/clone_rate_bound_certificate.py	passed	Run log
research/five_state_large_advantage_certificate.py	passed	Run log
research/thermal_ratio_obstruction_certificate.py	passed	Run log
research/reset_trace_obstruction.py	passed	Run log
research/reset_trace_independent_audit.py	passed	Run log
research/kl_contrast_gradient_obstruction.py	passed	Run log
research/test_exact_commensurate_classifier.py	passed	Run log
research/inverse_realizability_audit.py	passed	Run log
research/five_state_degenerate_nonmonotone.py	passed	Run log

Script	Run status	Evidence
research/extended_mode_obstruction_certificate.py	passed	Run log
research/independent_rational_classifier_audit.py	passed	Run log
research/independent_classifier_nearzero_audit.py	passed	Run log
research/exact_classifier_cross_audit.py	passed	Run log
research/check_entropy_reductions.py	passed	Run log
research/finite_time_entropy_independent_audit.py	passed	Run log
research/thermal_ratio_obstruction_certificate.py	passed	Run log
research/reset_trace_obstruction.py	passed	Run log
research/reset_trace_independent_audit.py	passed	Run log
research/finite_time_entropy_independent_audit.py	passed	Run log
research/five_state_extension_certificate.py	passed	Run log
research/high_dimension_amplitude_certificate.py	passed	Run log
research/clone_rate_bound_certificate.py	passed	Run log
research/five_state_large_advantage_certificate.py	passed	Run log
research/run_robust_finite_time_certificate.py	passed	Run log
research/certify_physical_nominal.py	passed	Run log
research/thermal_collision_model.py	passed	Run log
research/activated_intermediate_model.py	passed	Run log
research/check_activated_intermediates.py	passed	Run log

Script	Run status	Evidence
research/physics_plots.py	passed	Run log
research/mechanism_plots.py	passed	Run log
research/quarter_limit_plots.py	passed	Run log
research/physical_rate_plots.py	passed	Run log
research/focused_overview.py	passed	Run log
research/check_physical_rate_model.py	passed	Run log
research/check_figure_data.py	passed	Run log
research/check_pedagogical_revision.py	passed	Run log
research/audit_physical_goal_completion.py	passed	Run log
research/kl_contrast_gradient_obstruction.py	passed	Run log
research/check_entropy_reductions.py	passed	Run log
research/build_freshman_guide.py	passed	Run log
research/search_physical_rate_model.py	passed	Run log
research/search_physical_rate_model.py	passed	Run log
research/refine_physical_rate_model.py	passed	Run log
research/adversarial_physical_robustness.py	passed	Run log
research/build_papers.sh	passed	Run log
research/physics_plots.py	passed	Run log
research/mechanism_plots.py	passed	Run log

Script	Run status	Evidence
research/quarter_limit_plots.py	passed	Run log
research/physical_rate_plots.py	passed	Run log
research/focused_overview.py	passed	Run log
research/revise_report.py	passed	Run log
research/exact_classifier_cross_audit.py	passed	Run log
formal/check.sh	passed	Run log
research/revise_report.py	passed	Run log
research/check_pedagogical_revision.py	passed	Run log
research/check_publication_consistency.py	passed	Run log
research/check_historical_candidates.py	passed	Run log
research/audit_physical_goal_completion.py	passed	Run log
research/build_freshman_guide.py	passed	Run log
research/build_papers.sh	passed	Run log
research/cone_search.py	passed	Run log
research/adversarial_search.py	passed	Run log
research/adversarial_signed.py	passed	Run log
research/exact_mode.py	passed	Run log
research/exact_mode_variant.py	passed	Run log
research/general_n_arithmetic_highprecision.py	passed	Run log

Script	Run status	Evidence
research/general_n_inverse_candidate_recheck.py	passed	Run log
research/general_n_inverse_dual_search.py	passed	Run log
research/check_physical_rate_model.py	passed	Run log
research/check_physical_rate_model.py	passed	Run log
research/check_freshman_walkthrough.py	setup failure; corrected retry passed	Run log
research/test_freshman_walkthrough.py	passed	Run log
research/exact_mode_realizability.py	passed	Run log
research/classifier_audit.py	passed	Run log
research/ordered_conductance_numeric_audit.py	passed	Run log
research/exact_commensurate_classifier.py	passed	Run log
research/exact_rational_triplet_classifier.py	passed	Run log
research/check_transient_heating_candidate.py	passed	Run log
research/test_polynomial_reference.py	passed	Run log
research/test_entropy_audit_verdict.py	passed	Run log
research/test_robust_runner_inputs.py	passed	Run log
research/check_freshman_walkthrough.py	passed	Run log
research/standalone_primary_check.py	passed	Run log
research/test_standalone_primary_check.py	passed	Run log
research/check_temperature_reduction.cpp	passed	Run log

Script	Run status	Evidence
research/check_temperature_reduction.py	passed	Run log
research/check_threshold_constraint_logic.cpp	passed	Run log
./tmp/check_threshold_constraint_logic	passed	Run log
research/check_interval_primitives.cpp	passed	Run log
./tmp/check_interval_primitives	passed	Run log
research/check_quarter_approach.py	passed	Run log
research/check_publication_consistency.py	passed	Run log
research/check_sharpness_correspondence.py	passed	Run log

“Running” is a saved snapshot, not a live process check. Use the active terminal session to establish whether a process is still running. A passed replay is narrower than a completed proof review.

Numbered-claim audit

The same result may appear in both the report and appendices. Those duplicates are retained to expose document drift. Some statements already have strong formal or executable support; this ledger deliberately requires an explicit source-by-source proof review before marking an item complete.

Source location	Statement	Detailed review
focused_paper.tex line 105	Exact branch reduction	reviewed: stated scope
focused_paper.tex line 145	Strict triplet failure with ordinary cooling	reviewed: stated scope
focused_paper.tex line 368	Sharp amplitude bound	reviewed: stated scope

Source location	Statement	Detailed review
focused_paper.tex line 385	First-three mode restriction	reviewed: stated scope
research_report.tex line 2571	The branch logic reduces to a convex cone	reviewed: stated scope
research_report.tex line 2650	Counterexample to the published finite-temperature inference	reviewed: stated scope
research_report.tex line 2739	A single-turn criterion for a nearly neutral top state	reviewed: stated scope
research_report.tex line 2793	A controlled triplet-failing family with a sharp cooling threshold	reviewed: stated scope
research_report.tex line 2954	Application of stochastic monotonicity	reviewed: stated scope
research_report.tex line 2975	Untitled corollary	reviewed: stated scope
research_report.tex line 3009	Application: all-time convex-divergence ordering	reviewed: stated scope
research_report.tex line 3126	No strong effect and no anomalous heating	reviewed: stated scope
research_report.tex line 3291	Heating order for a tridiagonal contrast generator	reviewed: stated scope
research_report.tex line 3474	A bath-neighborhood KL ordering window	reviewed: stated scope
research_report.tex line 3590	Uniform cooling threshold	reviewed: stated scope

Source location	Statement	Detailed review
research_report.tex line 3675	A spacing-independent energy-span exclusion	reviewed: stated scope
research_report.tex line 3724	Amplitude bound for the contrast cone	reviewed: stated scope
research_report.tex line 3805	Strictly convex derivative curve	reviewed: stated scope
research_report.tex line 3870	Necessary and sufficient finite test	reviewed: stated scope
research_report.tex line 4037	State count and slow-space dimension	reviewed: stated scope
research_report.tex line 4216	Finite-time weighted-Euclidean classification	reviewed: stated scope
research_report.tex line 4336	Exact inverse feasibility	reviewed: stated scope
research_report.tex line 4431	A quarter-amplitude limit	reviewed: stated scope
research_report.tex line 4677	First-three arithmetic restriction	reviewed: stated scope
research_report.tex line 4807	Sharp quarter-amplitude bound	reviewed: stated scope
research_report.tex line 4861	Finite-span cooling bound	reviewed: stated scope
research_report.tex line 4940	Cooling is decided by one endpoint covariance	reviewed: stated scope

Source location	Statement	Detailed review
research_report.tex line 5037	A single faster mode must mix an adjacent pair	reviewed: stated scope
research_report.tex line 5149	The complete four-state degenerate boundary	reviewed: stated scope
research_report.tex line 5229	Rate evaluation of the triplet endpoint test	reviewed: stated scope
research_report.tex line 5282	Rational-input endpoint decision	reviewed: stated scope
research_report.tex line 5424	Dimension extension	reviewed: stated scope
research_report.tex line 5502	A dimension-dependent achievable advantage	reviewed: stated scope
research_report.tex line 5704	Conditional three-state reduction	reviewed: stated scope
research_report.tex line 5817	A constant rare-tail subfamily	reviewed: stated scope
research_report.tex line 5928	Graph-coupling feasibility for the arithmetic conjecture	reviewed: stated scope
focused_appendices.tex line 77	The branch logic reduces to a convex cone	reviewed: stated scope
focused_appendices.tex line 156	Counterexample to the published finite-temperature inference	reviewed: stated scope

Source location	Statement	Detailed review
focused_appendices.tex line 245	A single-turn criterion for a nearly neutral top state	reviewed: stated scope
focused_appendices.tex line 299	A controlled triplet-failing family with a sharp cooling threshold	reviewed: stated scope
focused_appendices.tex line 460	Application of stochastic monotonicity	reviewed: stated scope
focused_appendices.tex line 481	Untitled corollary	reviewed: stated scope
focused_appendices.tex line 515	Application: all-time convex-divergence ordering	reviewed: stated scope
focused_appendices.tex line 632	No strong effect and no anomalous heating	reviewed: stated scope
focused_appendices.tex line 797	Heating order for a tridiagonal contrast generator	reviewed: stated scope
focused_appendices.tex line 980	A bath-neighborhood KL ordering window	reviewed: stated scope
focused_appendices.tex line 1096	Uniform cooling threshold	reviewed: stated scope

Source location	Statement	Detailed review
focused_appendices.tex line 1181	A spacing-independent energy-span exclusion	reviewed: stated scope
focused_appendices.tex line 1230	Amplitude bound for the contrast cone	reviewed: stated scope
focused_appendices.tex line 1311	Strictly convex derivative curve	reviewed: stated scope
focused_appendices.tex line 1376	Necessary and sufficient finite test	reviewed: stated scope
focused_appendices.tex line 1543	State count and slow-space dimension	reviewed: stated scope
focused_appendices.tex line 1722	Finite-time weighted-Euclidean classification	reviewed: stated scope
focused_appendices.tex line 1842	Exact inverse feasibility	reviewed: stated scope
focused_appendices.tex line 1937	A quarter-amplitude limit	reviewed: stated scope
focused_appendices.tex line 2183	First-three arithmetic restriction	reviewed: stated scope

Source location	Statement	Detailed review
focused_appendices.tex line 2313	Sharp quarter-amplitude bound	reviewed: stated scope
focused_appendices.tex line 2367	Finite-span cooling bound	reviewed: stated scope
focused_appendices.tex line 2446	Cooling is decided by one endpoint covariance	reviewed: stated scope
focused_appendices.tex line 2543	A single faster mode must mix an adjacent pair	reviewed: stated scope
focused_appendices.tex line 2655	The complete four-state degenerate boundary	reviewed: stated scope
focused_appendices.tex line 2735	Rate evaluation of the triplet endpoint test	reviewed: stated scope
focused_appendices.tex line 2788	Rational-input endpoint decision	reviewed: stated scope
focused_appendices.tex line 2930	Dimension extension	reviewed: stated scope
focused_appendices.tex line 3008	A dimension-dependent achievable advantage	reviewed: stated scope

Source location	Statement	Detailed review
focused_appendices.tex line 3210	Conditional three-state reduction	reviewed: stated scope
focused_appendices.tex line 3323	A constant rare-tail subfamily	reviewed: stated scope
focused_appendices.tex line 3434	Graph-coupling feasibility for the arithmetic conjecture	reviewed: stated scope

The full scope inventory also lists source hashes, advertised commands, and unnumbered review obligations. The review notes record the checks actually performed and issues found.

What remains open in the science: the sharp higher-dimensional bound, general finite-time classifications, publication priority, and experimental validation. These are distinct from checking that the report describes its results correctly. The source-role index distinguishes current proof material, teaching material, fixed-version references, historical notes, and exploratory utilities. A file being listed is not a claim that every unused program is correct.

▼ **Final self-check: say the result without exaggeration**

"A valid four-state thermal model can fail every specified original-rate triplet test yet show ordinary anomalous cooling. Under the full four-state premise with a simple slow mode, the slow-amplitude advantage has sharp supremum 25%. That is not a universal cooling-time gain. A separate idealized physical model has continuous finite-time KL certificates on a stated parameter box. These results do not establish laboratory performance or resolve every higher-dimensional question."